Continuous Probability Distributions

7

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CRUISE SHIPS of the Royal Viking line report that 80% of their rooms are occupied during September. For a cruise ship having 800 rooms, what is the probability that 665 or more are occupied in September? (See Exercise 60 and LO7-4.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

LO7-1 Describe the uniform probability distribution and use it to calculate probabilities.

LO7-2 Describe the characteristics of a normal probability distribution.

LO7-3 Describe the standard normal probability distribution and use it to calculate probabilities.

LO7-4 Approximate the binomial probability distribution using the standard normal probability distribution to calculate probabilities.

LO7-5 Describe the exponential probability distribution and use it to calculate probabilities.

INTRODUCTION

Chapter 6 began our study of probability distributions. We consider three *discrete* probability distributions: binomial, hypergeometric, and Poisson. These distributions are based on discrete random variables, which can assume only clearly separated values. For example, we select for study 10 small businesses that began operations during the year 2014. The number still operating in 2017 can be 0, 1, 2, . . . , 10. There cannot be 3.7, 12, or −7 still operating in 2017. In this example, only certain outcomes are possible and these outcomes are represented by clearly separated values. In addition, the result is usually found by counting the number of successes. We count the number of the businesses in the study that are still in operation in 2017.

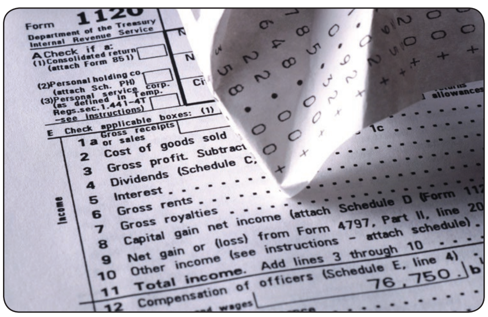
We continue our study of probability distributions by examining *continuous* probability distributions. A continuous probability distribution usually results from measuring something, such as the distance from the dormitory to the classroom, the weight of an individual, or the amount of bonus earned by CEOs. As an example, at Dave's Inlet Fish Shack flounder is the featured, fresh-fish menu item. The distribution of the amount of flounder sold per day has a mean of 10.0 pounds per day and a standard deviation of 3.0 pounds per day. This distribution is continuous because Dave, the owner, "measures" the amount of flounder sold each day. It is important to realize that a continuous random variable has an infinite number of values within a particular range. So, for a continuous random variable, probability is for a range of values. The probability for a specific value of a continuous random variable is 0.

This chapter shows how to use three continuous probability distributions: the uniform probability distribution, the normal probability distribution, and the exponential probability distribution.

LO7-1
Describe the uniform probability distribution and use it to calculate probabilities.

THE FAMILY OF UNIFORM PROBABILITY DISTRIBUTIONS

The uniform probability distribution is the simplest distribution for a continuous random variable. This distribution is rectangular in shape and is completely defined by its minimum and maximum values. Here are some examples that follow a uniform distribution.



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- The sales of gasoline at the Kwik Fill in Medina, New York, follow a uniform distribution that varies between 2,000 and 5,000 gallons per day. The random variable is the number of gallons sold per day and is continuous within the interval between 2,000 gallons and 5,000 gallons.
- Volunteers at the Grand Strand Public Library prepare federal income tax forms. The time to prepare form 1040-EZ follows a uniform distribution over the interval between 10 minutes and 30 minutes. The random variable is the number of minutes to complete the form, and it can assume any value between 10 and 30.

A uniform distribution is shown in Chart 7–1. The distribution's shape is rectangular and has a minimum value of a and a maximum of b. Also notice in Chart 7–1 the height of the distribution is constant or uniform for all values between a and b.

The mean of a uniform distribution is located in the middle of the interval between the minimum and maximum values. It is computed as:

MEAN OF THE UNIFORM DISTRIBUTION $$\mu = \frac{a+b}{2} \qquad \textbf{(7–1)}$$

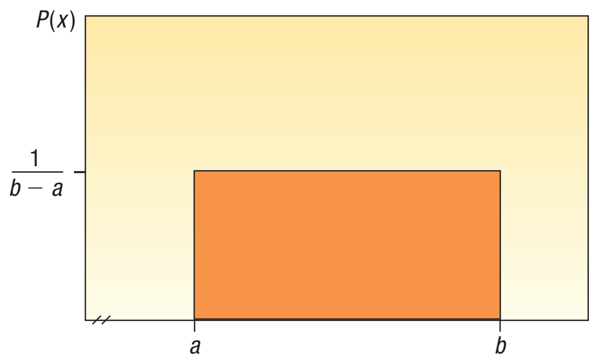


CHART 7–1 A Continuous Uniform Distribution

The standard deviation describes the dispersion of a distribution. In the uniform distribution, the standard deviation is also related to the interval between the maximum and minimum values.

STANDARD DEVIATION OF THE UNIFORM DISTRIBUTION

$$\sigma = \sqrt{\frac{(b-a)^2}{12}} \tag{7–2}$$

The equation for the uniform probability distribution is:

UNIFORM DISTRIBUTION

$$P(x) = \frac{1}{b-a} \text{ if } a \le x \le b \text{ and 0 elsewhere} \tag{7–3}$$

As we described in Chapter 6, probability distributions are useful for making probability statements concerning the values of a random variable. For distributions describing a continuous random variable, areas within the distribution represent probabilities. In the uniform distribution, its rectangular shape allows us to apply the area formula for a rectangle. Recall that we find the area of a rectangle by multiplying its length by its height. For the uniform distribution, the height of the rectangle is $P(x)$, which is $1/(b - a)$. The length or base of the distribution is $b - a$. So if we multiply the height of the distribution by its entire range to find the area, the result is always 1.00. To put it another way, the total area within a continuous probability distribution is equal to 1.00. In general

$$\text{Area} = (\text{height})(\text{base}) = \frac{1}{(b-a)}(b-a) = 1.00$$

So if a uniform distribution ranges from 10 to 15, the height is 0.20, found by 1/(15 − 10). The base is 5, found by 15 − 10. The total area is:

$$\text{Area} = (\text{height})(\text{base}) = \frac{1}{(15-10)}(15-10) = 1.00$$

The following example illustrates the features of a uniform distribution and how we use it to calculate probabilities.

EXAMPLE

Southwest Arizona State University provides bus service to students while they are on campus. A bus arrives at the North Main Street and College Drive stop every 30 minutes between 6 a.m. and 11 p.m. during weekdays. Students arrive at the bus stop at random times. The time that a student waits is uniformly distributed from 0 to 30 minutes.

1. Draw a graph of this distribution.
2. Show that the area of this uniform distribution is 1.00.
3. How long will a student "typically" have to wait for a bus? In other words, what is the mean waiting time? What is the standard deviation of the waiting times?
4. What is the probability a student will wait more than 25 minutes?
5. What is the probability a student will wait between 10 and 20 minutes?

SOLUTION

In this case, the random variable is the length of time a student must wait. Time is measured on a continuous scale, and the wait times may range from 0 minutes up to 30 minutes.

1. The graph of the uniform distribution is shown in Chart 7–2. The horizontal line is drawn at a height of .0333, found by 1/(30 – 0). The range of this distribution is 30 minutes.

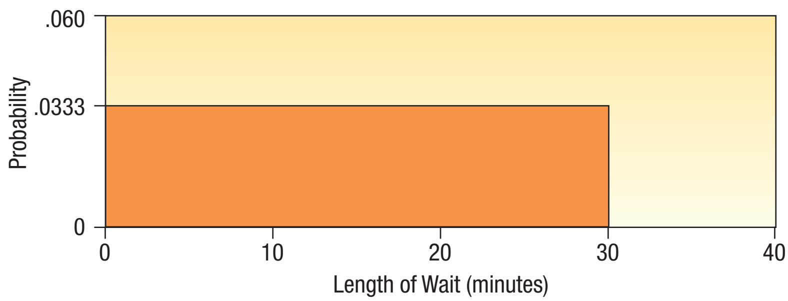


CHART 7–2 Uniform Probability Distribution of Student Waiting Times

2. The times students must wait for the bus are uniform over the interval from 0 minutes to 30 minutes, so in this case a is 0 and b is 30.

$$\text{Area} = (\text{height})(\text{base}) = \frac{1}{(30-0)}(30-0) = 1.00$$

3. To find the mean, we use formula (7–1).

$$\mu = \frac{a+b}{2} = \frac{0+30}{2} = 15$$

The mean of the distribution is 15 minutes, so the typical wait time for bus service is 15 minutes.
To find the standard deviation of the wait times, we use formula (7–2).

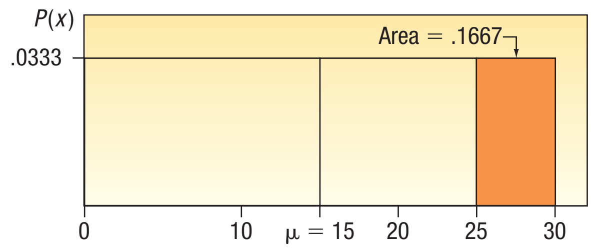
$$\sigma = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{(30-0)^2}{12}} = 8.66$$

The standard deviation of the distribution is 8.66 minutes. This measures the variation in the student wait times.

4. The area within the distribution for the interval 25 to 30 represents this particular probability. From the area formula:

$$P(25 < \text{wait time} < 30) = (\text{height})(\text{base}) = \frac{1}{(30-0)}(5) = .1667$$

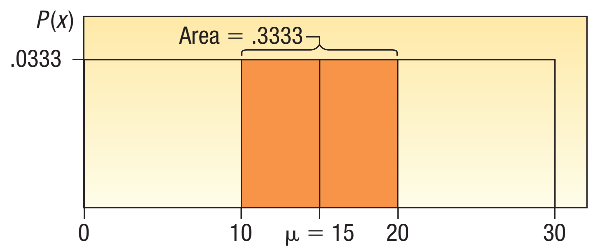
So the probability a student waits between 25 and 30 minutes is .1667. This conclusion is illustrated by the following graph.



5. The area within the distribution for the interval 10 to 20 represents the probability.

$$P(10 < \text{wait time} < 20) = (\text{height})(\text{base}) = \frac{1}{(30 - 0)}(10) = .3333$$

We can illustrate this probability as follows.



SELF-REVIEW 7–1

Microwave ovens only last so long. The life-time of a microwave oven follows a uniform distribution between 8 and 14 years.
(a) Draw this uniform distribution. What are the height and base values?
(b) Show the total area under the curve is 1.00.
(c) Calculate the mean and the standard deviation of this distribution.
(d) What is the probability a particular microwave oven lasts between 10 and 14 years?
(e) What is the probability a microwave oven will last less than 9 years?

EXERCISES

1. A uniform distribution is defined over the interval from 6 to 10.
- **a.** What are the values for *a* and *b*?
- **b.** What is the mean of this uniform distribution?
- **c.** What is the standard deviation?
- **d.** Show that the total area is 1.00.
- **e.** Find the probability of a value more than 7.
- **f.** Find the probability of a value between 7 and 9.

2. A uniform distribution is defined over the interval from 2 to 5.
- **a.** What are the values for *a* and *b*?
- **b.** What is the mean of this uniform distribution?
- **c.** What is the standard deviation?
- **d.** Show that the total area is 1.00.
- **e.** Find the probability of a value more than 2.6.
- **f.** Find the probability of a value between 2.9 and 3.7.

3. The closing price of Schnur Sporting Goods Inc. common stock is uniformly distributed between \$20 and \$30 per share. What is the probability that the stock price will be:
   a. More than \$27?
   b. Less than or equal to \$24?
4. According to the Insurance Institute of America, a family of four spends between \$400 and \$3,800 per year on all types of insurance. Suppose the money spent is uniformly distributed between these amounts.
   a. What is the mean amount spent on insurance?
   b. What is the standard deviation of the amount spent?
   c. If we select a family at random, what is the probability they spend less than \$2,000 per year on insurance per year?
   d. What is the probability a family spends more than \$3,000 per year?
5. The April rainfall in Flagstaff, Arizona, follows a uniform distribution between 0.5 and 3.00 inches.
   a. What are the values for *a* and *b*?
   b. What is the mean amount of rainfall for the month? What is the standard deviation?
   c. What is the probability of less than an inch of rain for the month?
   d. What is the probability of *exactly* 1.00 inch of rain?
   e. What is the probability of more than 1.50 inches of rain for the month?
6. Customers experiencing technical difficulty with their Internet cable service may call an 800 number for technical support. It takes the technician between 30 seconds and 10 minutes to resolve the problem. The distribution of this support time follows the uniform distribution.
   a. What are the values for *a* and *b* in minutes?
   b. What is the mean time to resolve the problem? What is the standard deviation of the time?
   c. What percent of the problems take more than 5 minutes to resolve?
   d. Suppose we wish to find the middle 50% of the problem-solving times. What are the end points of these two times?

LO7-2
Describe the characteristics of a normal probability distribution.

THE FAMILY OF NORMAL PROBABILITY DISTRIBUTIONS

Next we consider the normal probability distribution. Unlike the uniform distribution [see formula (7–3)] the normal probability distribution has a very complex formula.

NORMAL PROBABILITY DISTRIBUTION

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\left[\frac{(x-\mu)^2}{2\sigma^2}\right]} \qquad \textbf{(7–4)}$$

However, do not be bothered by how complex this formula looks. You are already familiar with many of the values. The symbols μ and σ refer to the mean and the standard deviation, as usual. The Greek symbol π is a constant and its value is approximately 22/7 or 3.1416. The letter *e* is also a constant. It is the base of the natural log system and is approximately equal to 2.718. *x* is the value of a continuous random variable. So a normal distribution is based on—that is, it is defined by—its mean and standard deviation.

You will not need to make calculations using formula (7–4). Instead you will use a table, given in Appendix B.3, to find various probabilities. These probabilities can also be calculated using Excel functions as well as other statistical software.

The normal probability distribution has the following characteristics:

- It is **bell-shaped** and has a single peak at the center of the distribution. The arithmetic mean, median, and mode are equal and located in the center of the distribution. The total area under the curve is 1.00. Half the area under the normal curve is to the right of this center point and the other half, to the left of it.
- It is **symmetrical** about the mean. If we cut the normal curve vertically at the center value, the shapes of the curves will be mirror images. Also, the area of each half is 0.5.
- It falls off smoothly in either direction from the central value. That is, the distribution is **asymptotic:** The curve gets closer and closer to the *X*-axis but never actually touches it. To put it another way, the tails of the curve extend indefinitely in both directions.
- The location of a normal distribution is determined by the mean, μ. The dispersion or spread of the distribution is determined by the standard deviation, σ.

These characteristics are shown graphically in Chart 7–3.

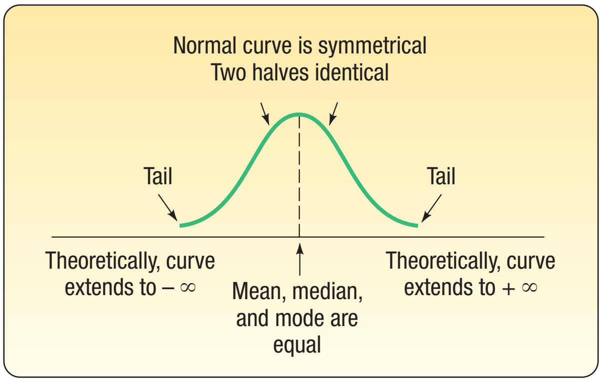


CHART 7–3 Characteristics of a Normal Distribution

There is not just one normal probability distribution, but rather a "family" of them. For example, in Chart 7–4 the probability distributions of length of employee service in three different plants are compared. In the Camden plant, the mean is 20 years and the standard deviation is 3.1 years. There is another normal probability distribution for the length of service in the Dunkirk plant, where $\mu = 20$ years and $\sigma = 3.9$ years. In the Elmira plant, $\mu = 20$ years and $\sigma = 5.0$ years. Note that the means are the same but the standard deviations are different. As the standard deviation gets smaller, the distribution becomes more narrow and "peaked."

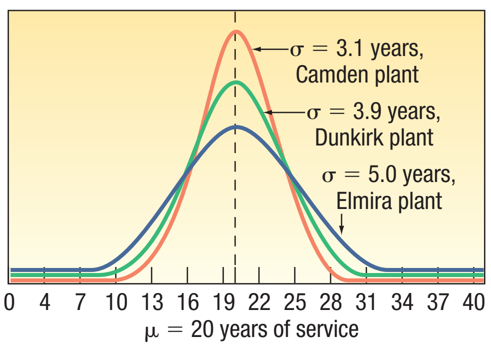


CHART 7–4 Normal Probability Distributions with Equal Means but Different Standard Deviations

STATISTICS IN ACTION

Many variables are approximately, normally distributed, such as IQ scores, life expectancies, and adult height. This implies that nearly all observations occur within 3 standard deviations of the mean. On the other hand, observations that occur beyond 3 standard deviations from the mean are extremely rare. For example, the mean adult male height is 68.2 inches (about 5 feet 8 inches) with a standard deviation of 2.74. This means that almost all males are between 60.0 inches (5 feet) and 76.4 inches (6 feet 4 inches). LeBron James, a professional basketball player with the Cleveland Cavaliers, is 80 inches, or 6 feet 8 inches, which is clearly beyond 3 standard deviations from the mean. The height of a standard doorway is 6 feet 8 inches, and should be high enough for almost all adult males, except for a rare person like LeBron James.

As another example, the driver's seat in most vehicles is set to comfortably fit a person who is at least 159 cm (62.5 inches) tall. The distribution of heights of adult women is approximately a normal distribution with a mean of 161.5 cm and a standard deviation of 6.3 cm. Thus about 35% of adult women will not fit comfortably in the driver's seat.

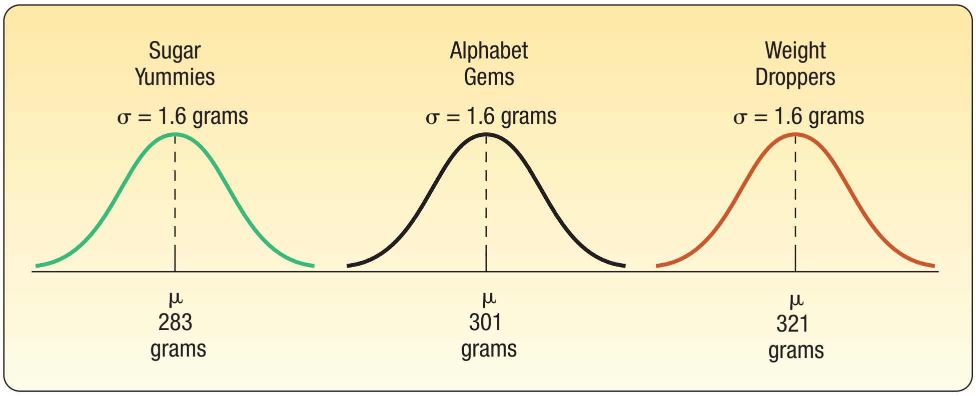


CHART 7–5 Normal Probability Distributions Having Different Means but Equal Standard Deviations

Chart 7–5 shows the distribution of box weights of three different cereals. The weights follow a normal distribution with different means but identical standard deviations.

Finally, Chart 7–6 shows three normal distributions having different means and standard deviations. They show the distribution of tensile strengths, measured in pounds per square inch (psi), for three types of cables.

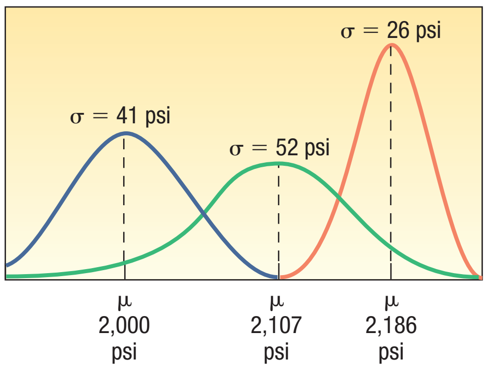


CHART 7–6 Normal Probability Distributions with Different Means and Standard Deviations

In Chapter 6, recall that discrete probability distributions show the specific likelihood a discrete value will occur. For example, on page 186 the binomial distribution is used to calculate the probability that none of the five flights arriving at the Bradford Pennsylvania Regional Airport will be late.

With a continuous probability distribution, areas below the curve define probabilities. The total area under the normal curve is 1.0. This accounts for all possible outcomes. Because a normal probability distribution is symmetric, the area under the curve to the left of the mean is 0.5, and the area under the curve to the right of the mean is 0.5. Apply this to the distribution of Sugar Yummies in Chart 7–5. It is normally distributed with a mean of 283 grams. Therefore, the probability of filling a box with more than 283 grams is 0.5 and the probability of filling a box with less than 283 grams is 0.5. We also can determine the probability that a box weighs between 280 and 286 grams. However, to determine this probability we need to know about the standard normal probability distribution.

LO7-3
Describe the standard normal probability distribution and use it to calculate probabilities.

THE STANDARD NORMAL PROBABILITY DISTRIBUTION

The number of normal distributions is unlimited, each having a different mean (μ), standard deviation (σ), or both. While it is possible to provide a limited number of probability tables for discrete distributions such as the binomial and the Poisson, providing tables for the infinite number of normal distributions is impractial. Fortunately, one member of the family can be used to determine the probabilities for all normal probability distributions. It is called the **standard normal probability distribution,** and it is unique because it has a mean of 0 and a standard deviation of 1.

Any *normal probability distribution* can be converted into a *standard normal probability distribution* by subtracting the mean from each observation and dividing this difference by the standard deviation. The results are called ***z* values** or ***z* scores.**

***z* VALUE** The signed distance between a selected value, designated *x,* and the mean, μ, divided by the standard deviation, σ.

So, a z value is the distance from the mean, measured in units of the standard deviation. The formula for this conversion is:

STANDARD NORMAL VALUE $$z = \frac{x - \mu}{\sigma} \quad \textbf{(7–5)}$$

where:

x is the value of any particular observation or measurement.
μ is the mean of the distribution.
σ is the standard deviation of the distribution.

STATISTICS IN ACTION

An individual's skills depend on a combination of many hereditary and environmental factors, each having about the same amount of weight or influence on the skills. Thus, much like a binomial distribution with a large number of trials, many skills and attributes follow the normal distribution. For example, the SAT Reasoning Test is the most widely used standardized test for college admissions in the United States. Scores are based on a normal distribution with a mean of 1,500 and a standard deviation of 300.

As we noted in the preceding definition, a z value expresses the distance or difference between a particular value of x and the arithmetic mean in units of the standard deviation. Once the normally distributed observations are standardized, the z values are normally distributed with a mean of 0 and a standard deviation of 1. Therefore, the z distribution has all the characteristics of any normal probability distribution. These characteristics are listed on page 215 in the Family of Normal Probability Distributions section. The table in Appendix B.3 lists the probabilities for the standard normal probability distribution. A small portion of this table follows.

TABLE 7–1 Areas under the Normal Curve

z	0.00	0.01	0.02	0.03	0.04	0.05	. . .
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	
⋮							

Applications of the Standard Normal Distribution

The standard normal distribution is very useful for determining probabilities for any normally distributed random variable. The basic procedure is to find the z value for a particular value of the random variable based on the mean and standard deviation of its distribution. Then, using the z value, we can use the standard normal distribution to find various probabilities. The following example/solution describes the details of the application.

EXAMPLE

In recent years a new type of taxi service has evolved in more than 300 cities worldwide, where the customer is connected directly with a driver via a smartphone. The idea was first developed by Uber Technologies, which is headquartered in San Francisco, California. It uses the Uber mobile app, which allows customers with a smartphone to submit a trip request which is then routed to a Uber driver who picks up the customer and takes the customer to the desired location. No cash is involved, the payment for the transaction is handled via a digital payment.

Suppose the weekly income of Uber drivers follows the normal probability distribution with a mean of \$1,000 and a standard deviation of \$100. What is the z value of income for a driver who earns \$1,100 per week? For a driver who earns \$900 per week?

SOLUTION

Using formula (7–5), the z values corresponding to the two x values (\$1,100 and \$900) are:

For $x = \$1{,}100$:

$$z = \frac{x - \mu}{\sigma} = \frac{\$1{,}100 - \$1{,}000}{\$100} = 1.00$$

For $x = \$900$:

$$z = \frac{x - \mu}{\sigma} = \frac{\$900 - \$1{,}000}{\$100} = -1.00$$

The z of 1.00 indicates that a weekly income of \$1,100 is one standard deviation above the mean, and a z of -1.00 shows that a \$900 income is one standard deviation below the mean. Note that both incomes (\$1,100 and \$900) are the same distance (\$100) from the mean.

SELF-REVIEW 7–2

A recent national survey concluded that the typical person consumes 48 ounces of water per day. Assume daily water consumption follows a normal probability distribution with a standard deviation of 12.8 ounces.

(a) What is the z value for a person who consumes 64 ounces of water per day? Based on this z value, how does this person compare to the national average?

(b) What is the z value for a person who consumes 32 ounces of water per day? Based on this z value, how does this person compare to the national average?

The Empirical Rule

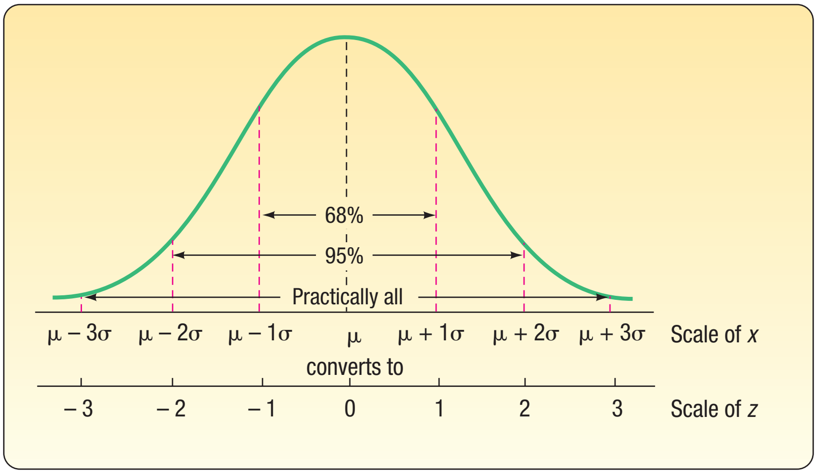
The Empirical Rule is introduced on page 80 of Chapter 3. It states that if a random variable is normally distributed, then:

1. Approximately 68% of the observations will lie within plus and minus one standard deviation of the mean.

2. About 95% of the observations will lie within plus and minus two standard deviations of the mean.
3. Practically all, or 99.7% of the observations, will lie within plus and minus three standard deviations of the mean.

Now, knowing how to apply the standard normal probability distribution, we can verify the Empirical Rule. For example, one standard deviation from the mean is the same as a z value of 1.00. When we refer to the standard normal probability table, a z value of 1.00 corresponds to a probability of 0.3413. So what percent of the observations will lie within plus and minus one standard deviation of the mean? We multiply (2)(0.3413), which equals 0.6826, or approximately 68% of the observations are within plus and minus one standard deviation of the mean.

The Empirical Rule is summarized in the following graph.



Transforming measurements to standard normal deviates changes the scale. The conversions are also shown in the graph. For example, $\mu + 1\sigma$ is converted to a z value of 1.00. Likewise, $\mu - 2\sigma$ is transformed to a z value of -2.00. Note that the center of the z distribution is zero, indicating no deviation from the mean, μ.

EXAMPLE

As part of its quality assurance program, the Autolite Battery Company conducts tests on battery life. For a particular D-cell alkaline battery, the mean life is 19 hours. The useful life of the battery follows a normal distribution with a standard deviation of 1.2 hours. Answer the following questions.

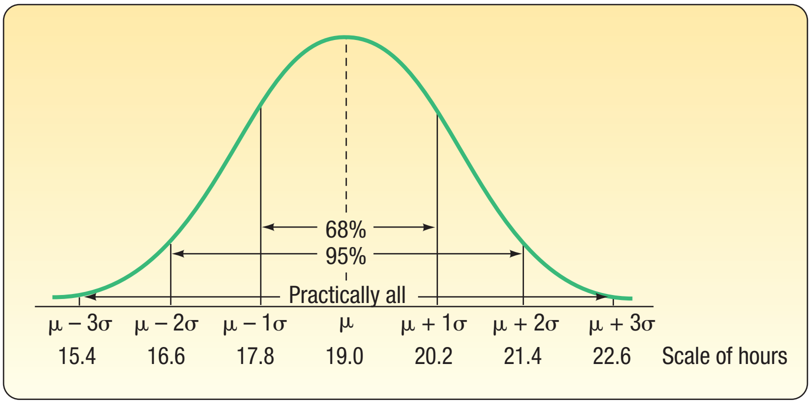
1. About 68% of the batteries failed between what two values?
2. About 95% of the batteries failed between what two values?
3. Virtually all of the batteries failed between what two values?

SOLUTION

We can use the Empirical Rule to answer these questions.

1. About 68% of the batteries will fail between 17.8 and 20.2 hours, found by $19.0 \pm 1(1.2)$ hours.
2. About 95% of the batteries will fail between 16.6 and 21.4 hours, found by $19.0 \pm 2(1.2)$ hours.
3. Practically all failed between 15.4 and 22.6 hours, found by $19.0 \pm 3(1.2)$ hours.

This information is summarized on the following chart.



SELF-REVIEW 7–3

The distribution of the annual incomes of a group of middle-management employees at Compton Plastics approximates a normal distribution with a mean of \$47,200 and a standard deviation of \$800.

(a) About 68% of the incomes lie between what two amounts?
(b) About 95% of the incomes lie between what two amounts?
(c) Virtually all of the incomes lie between what two amounts?
(d) What are the median and the modal incomes?
(e) Is the distribution of incomes symmetrical?

EXERCISES

7. Explain what is meant by this statement: "There is not just one normal probability distribution but a 'family' of them."

8. List the major characteristics of a normal probability distribution.

9. The mean of a normal probability distribution is 500; the standard deviation is 10.
- **a.** About 68% of the observations lie between what two values?
- **b.** About 95% of the observations lie between what two values?
- **c.** Practically all of the observations lie between what two values?

10. The mean of a normal probability distribution is 60; the standard deviation is 5.
- **a.** About what percent of the observations lie between 55 and 65?
- **b.** About what percent of the observations lie between 50 and 70?
- **c.** About what percent of the observations lie between 45 and 75?

11. The Kamp family has twins, Rob and Rachel. Both Rob and Rachel graduated from college 2 years ago, and each is now earning \$50,000 per year. Rachel works in the retail industry, where the mean salary for executives with less than 5 years' experience is \$35,000 with a standard deviation of \$8,000. Rob is an engineer. The mean salary for engineers with less than 5 years' experience is \$60,000 with a standard deviation of \$5,000. Compute the z values for both Rob and Rachel and comment on your findings.

12. A recent article in the *Cincinnati Enquirer* reported that the mean labor cost to repair a heat pump is \$90 with a standard deviation of \$22. Monte's Plumbing and Heating Service completed repairs on two heat pumps this morning. The labor cost for the first was \$75 and it was \$100 for the second. Assume the distribution of labor costs follows the normal probability distribution. Compute z values for each and comment on your findings.

Finding Areas under the Normal Curve

The next application of the standard normal distribution involves finding the area in a normal distribution between the mean and a selected value, which we identify as x. The following example/solution will illustrate the details.

EXAMPLE

In the first example/solution described on page 218 in this section, we reported that the weekly income of Uber drivers followed the normal distribution with a mean of \$1,000 and a standard deviation of \$100. That is, $\mu = \$1{,}000$ and $\sigma = \$100$. What is the likelihood of selecting a driver whose weekly income is between \$1,000 and \$1,100?

SOLUTION

We have already converted \$1,100 to a z value of 1.00 using formula (7–5). To repeat:

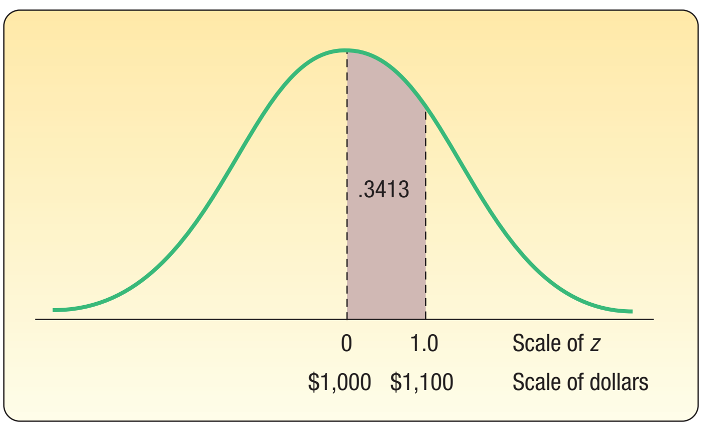
$$z = \frac{x - \mu}{\sigma} = \frac{\$1{,}100 - \$1{,}000}{\$100} = 1.00$$

The probability associated with a z of 1.00 is available in Appendix B.3. A portion of Appendix B.3 follows. To locate the probability, go down the left column to 1.0, and then move horizontally to the column headed .00. The value is .3413.

z	0.00	0.01	0.02
⋮	⋮	⋮	⋮
0.7	.2580	.2611	.2642
0.8	.2881	.2910	.2939
0.9	.3159	.3186	.3212
1.0	.3413	.3438	.3461
1.1	.3643	.3665	.3686
⋮	⋮	⋮	⋮

The area under the normal curve between \$1,000 and \$1,100 is .3413. We could also say 34.13% of Uber drivers earn between \$1,000 and \$1,100 weekly, or the likelihood of selecting a driver and finding his or her income is between \$1,000 and \$1,100 is .3413.

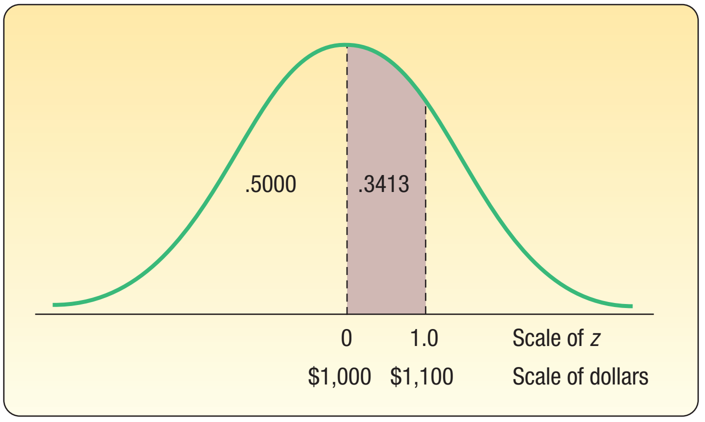
This information is summarized in the following diagram.



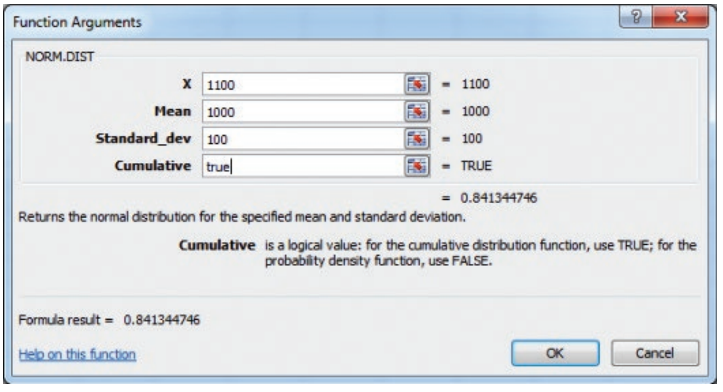
STATISTICS IN ACTION

Many processes, such as filling soda bottles and canning fruit, are normally distributed. Manufacturers must guard against both over- and underfilling. If they put too much in the can or bottle, they are giving away their product. If they put too little in, the customer may feel cheated and the government may question the label description. "Control charts," with limits drawn three standard deviations above and below the mean, are routinely used to monitor this type of production process.

In the example/solution just completed, we are interested in the probability between the mean and a given value. Let's change the question. Instead of wanting to know the probability of selecting a random driver who earned between $1,000 and $1,100, suppose we wanted the probability of selecting a driver who earned less than $1,100. In probability notation, we write this statement as *P*(weekly income < $1,100). The method of solution is the same. We find the probability of selecting a driver who earns between $1,000, the mean, and $1,100. This probability is .3413. Next, recall that half the area, or probability, is above the mean and half is below. So the probability of selecting a driver earning less than $1,000 is .5000. Finally, we add the two probabilities, so .3413 + .5000 = .8413. About 84% of Uber drivers earn less than $1,100 per week. See the following diagram.



Excel will calculate this probability. The necessary commands are in the **Software Commands** in Appendix C. The answer is .8413, the same as we calculated.



EXAMPLE

Refer to the first example/solution discussed on page 218 in this section regarding the weekly income of Uber drivers. The distribution of weekly incomes follows the normal probability distribution, with a mean of $1,000 and a standard deviation of $100. What is the probability of selecting a driver whose income is:

1. Between $790 and $1,000?
2. Less than $790?

SOLUTION

We begin by finding the z value corresponding to a weekly income of \$790. From formula (7–5):

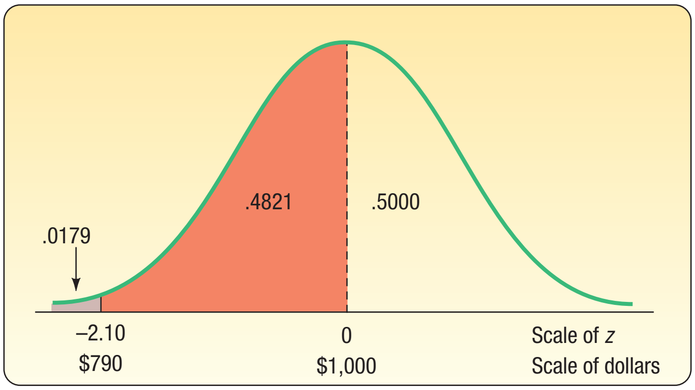
$$z = \frac{x - \mu}{s} = \frac{\$790 - \$1{,}000}{\$100} = -2.10$$

See Appendix B.3. Move down the left margin to the row 2.1 and across that row to the column headed 0.00. The value is .4821. So the area under the standard normal curve corresponding to a z value of 2.10 is .4821. However, because the normal distribution is symmetric, the area between 0 and a negative z value is the same as that between 0 and the corresponding positive z value. The likelihood of finding a driver earning between \$790 and \$1,000 is .4821. In probability notation, we write $P(\$790 < \text{weekly income} < \$1{,}000) = .4821$.

z	0.00	0.01	0.02
⋮	⋮	⋮	⋮
2.0	.4772	.4778	.4783
2.1	.4821	.4826	.4830
2.2	.4861	.4864	.4868
2.3	.4893	.4896	.4898
⋮	⋮	⋮	⋮

The mean divides the normal curve into two identical halves. The area under the half to the left of the mean is .5000, and the area to the right is also .5000. Because the area under the curve between \$790 and \$1,000 is .4821, the area below \$790 is .0179, found by .5000 − .4821. In probability notation, we write $P(\text{weekly income} < \$790) = .0179$.

So we conclude that 48.21% of the Uber drivers have weekly incomes between \$790 and \$1,000. Further, we can anticipate that 1.79% earn less than \$790 per week. This information is summarized in the following diagram.



SELF-REVIEW 7–4

The temperature of coffee sold at the Coffee Bean Cafe follows the normal probability distribution, with a mean of 150 degrees. The standard deviation of this distribution is 5 degrees.

(a) What is the probability that the coffee temperature is between 150 degrees and 154 degrees?
(b) What is the probability that the coffee temperature is more than 164 degrees?

EXERCISES

13. A normal population has a mean of 20.0 and a standard deviation of 4.0.
- **a.** Compute the z value associated with 25.0.
- **b.** What proportion of the population is between 20.0 and 25.0?
- **c.** What proportion of the population is less than 18.0?

14. A normal population has a mean of 12.2 and a standard deviation of 2.5.
- **a.** Compute the z value associated with 14.3.
- **b.** What proportion of the population is between 12.2 and 14.3?
- **c.** What proportion of the population is less than 10.0?

15. A recent study of the hourly wages of maintenance crew members for major airlines showed that the mean hourly salary was \$20.50, with a standard deviation of \$3.50. Assume the distribution of hourly wages follows the normal probability distribution. If we select a crew member at random, what is the probability the crew member earns:
- **a.** Between \$20.50 and \$24.00 per hour?
- **b.** More than \$24.00 per hour?
- **c.** Less than \$19.00 per hour?

16. The mean of a normal probability distribution is 400 pounds. The standard deviation is 10 pounds.
- **a.** What is the area between 415 pounds and the mean of 400 pounds?
- **b.** What is the area between the mean and 395 pounds?
- **c.** What is the probability of selecting a value at random and discovering that it has a value of less than 395 pounds?

Another application of the normal distribution involves combining two areas, or probabilities. One of the areas is to the right of the mean and the other to the left.

EXAMPLE

Continuing the example/solution first discussed on page 218 using the weekly income of Uber drivers, weekly income follows the normal probability distribution, with a mean of \$1,000 and a standard deviation of \$100. What is the area under this normal curve between \$840 and \$1,200?

SOLUTION

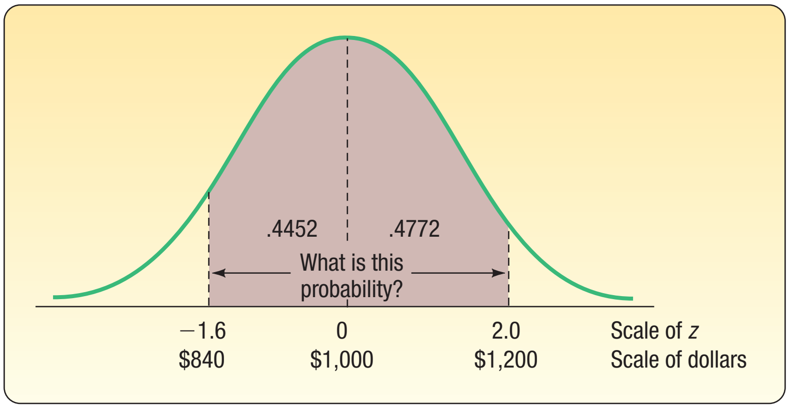
The problem can be divided into two parts. For the area between \$840 and the mean of \$1,000:

$$z = \frac{\$840 - \$1{,}000}{\$100} = \frac{-\$160}{\$100} = -1.60$$

For the area between the mean of \$1,000 and \$1,200:

$$z = \frac{\$1{,}200 - \$1{,}000}{\$100} = \frac{\$200}{\$100} = 2.00$$

The area under the curve for a z of -1.60 is .4452 (from Appendix B.3). The area under the curve for a z of 2.00 is .4772. Adding the two areas: $.4452 + .4772 = .9224$. Thus, the probability of selecting an income between \$840 and \$1,200 is .9224. In probability notation, we write $P(\$840 < \text{weekly income} < \$1{,}200) = .4452 + .4772 = .9224$. To summarize, 92.24% of the drivers have weekly incomes between \$840 and \$1,200. This is shown in a diagram:



Another application of the normal distribution involves determining the area between values on the *same* side of the mean.

EXAMPLE

Returning to the weekly income distribution of Uber drivers ($\mu = \$1,000$, $\sigma = \$100$), what is the area under the normal curve between \$1,150 and \$1,250?

SOLUTION

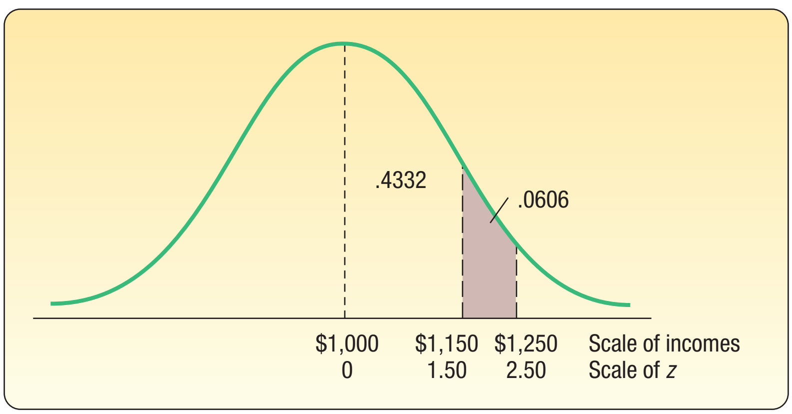
The situation is again separated into two parts, and formula (7–5) is used. First, we find the z value associated with a weekly income of \$1,250:

$$z = \frac{\$1,250 - \$1,000}{\$100} = 2.50$$

Next we find the z value for a weekly income of \$1,150:

$$z = \frac{\$1,150 - \$1,000}{\$100} = 1.50$$

From Appendix B.3, the area associated with a z value of 2.50 is .4938. So the probability of a weekly income between \$1,000 and \$1,250 is .4938. Similarly, the area associated with a z value of 1.50 is .4332, so the probability of a weekly income between \$1,000 and \$1,150 is .4332. The probability of a weekly income between \$1,150 and \$1,250 is found by subtracting the area associated with a z value of 1.50 (.4332) from that associated with a z of 2.50 (.4938). Thus, the probability of a weekly income between \$1,150 and \$1,250 is .0606. In probability notation, we write $P(\$1,150 < \text{weekly income} < \$1,250) = .4938 - .4332 = .0606$.



To summarize, there are four situations for finding the area under the standard normal probability distribution.

1. To find the area between 0 and z or $(-z)$, look up the probability directly in the table.
2. To find the area beyond z or $(-z)$, locate the probability of z in the table and subtract that probability from .5000.
3. To find the area between two points on different sides of the mean, determine the z values and add the corresponding probabilities.
4. To find the area between two points on the same side of the mean, determine the z values and subtract the smaller probability from the larger.

SELF-REVIEW 7–5

Refer to Self-Review 7–4. The temperature of coffee sold at the Coffee Bean Cafe follows the normal probability distribution with a mean of 150 degrees. The standard deviation of this distribution is 5 degrees.

(a) What is the probability the coffee temperature is between 146 degrees and 156 degrees?
(b) What is the probability the coffee temperature is more than 156 but less than 162 degrees?

EXERCISES

17. A normal distribution has a mean of 50 and a standard deviation of 4.
- **a.** Compute the probability of a value between 44.0 and 55.0.
- **b.** Compute the probability of a value greater than 55.0.
- **c.** Compute the probability of a value between 52.0 and 55.0.

18. A normal population has a mean of 80.0 and a standard deviation of 14.0.
- **a.** Compute the probability of a value between 75.0 and 90.0.
- **b.** Compute the probability of a value of 75.0 or less.
- **c.** Compute the probability of a value between 55.0 and 70.0.

19. Suppose the Internal Revenue Service reported that the mean tax refund for the year 2016 was $2,800. Assume the standard deviation is $450 and that the amounts refunded follow a normal probability distribution.
- **a.** What percent of the refunds are more than $3,100?
- **b.** What percent of the refunds are more than $3,100 but less than $3,500?
- **c.** What percent of the refunds are more than $2,250 but less than $3,500?

20. The distribution of the number of viewers for the *American Idol* television show follows a normal distribution with a mean of 29 million and a standard deviation of 5 million. What is the probability next week's show will:
- **a.** Have between 30 and 34 million viewers?
- **b.** Have at least 23 million viewers?
- **c.** Exceed 40 million viewers?

21. WNAE, an all-news AM station, finds that the distribution of the lengths of time listeners are tuned to the station follows the normal distribution. The mean of the distribution is 15.0 minutes and the standard deviation is 3.5 minutes. What is the probability that a particular listener will tune in for:
- **a.** More than 20 minutes?
- **b.** 20 minutes or less?
- **c.** Between 10 and 12 minutes?

22. Among the thirty largest U.S. cities, the mean one-way commute time to work is 25.8 minutes. https://deepblue.lib.umich.edu/bitstream/handle/2027.42/112057/103196.pdf?sequence=1&isAllowed=y. The longest one-way travel time is in New York City, where the mean time is 39.7 minutes. Assume the distribution of travel times in New York City follows the normal probability distribution and the standard deviation is 7.5 minutes.
- **a.** What percent of the New York City commutes are for less than 30 minutes?
- **b.** What percent are between 30 and 35 minutes?
- **c.** What percent are between 30 and 50 minutes?

The previous example/solutions require finding the percent of the observations located between two observations or the percent of the observations above, or below, a particular observation x. A further application of the normal distribution involves finding the value of the observation x when the percent above or below the observation is given.

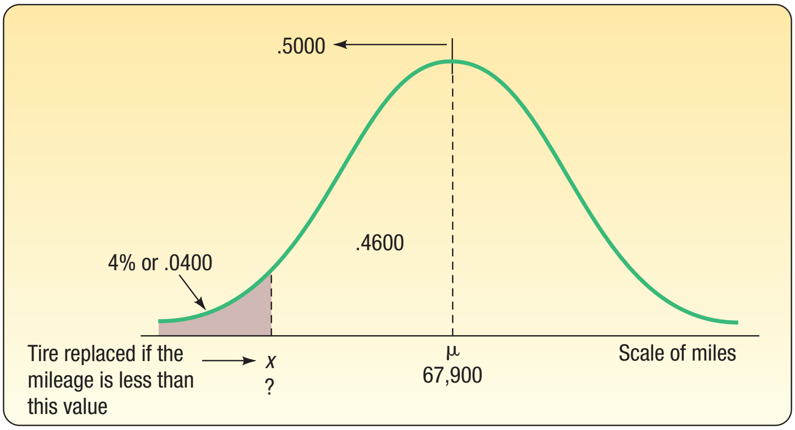
EXAMPLE

Layton Tire and Rubber Company wishes to set a minimum mileage guarantee on its new MX100 tire. Tests reveal the mean mileage is 67,900 with a standard deviation of 2,050 miles and that the distribution of miles follows the normal probability distribution. Layton wants to set the minimum guaranteed mileage so that no more than 4% of the tires will have to be replaced. What minimum guaranteed mileage should Layton announce?

© JupiterImages/Getty Images

SOLUTION

The facets of this case are shown in the following diagram, where x represents the minimum guaranteed mileage.



Inserting these values in formula (7–5) for z gives:

$$z = \frac{x - \mu}{\sigma} = \frac{x - 67{,}900}{2{,}050}$$

There are two unknowns in this equation, z and x. To find x, we first find z, and then solve for x. Recall from the characteristics of a normal curve that the area to the left of μ is .5000. The area between μ and x is .4600, found by .5000 − .0400. Now refer to Appendix B.3. Search the body of the table for the area closest to .4600. The closest area is .4599. Move to the margins from this value and read the z value of 1.75. Because the value is to the left of the mean, it is actually −1.75. These steps are illustrated in Table 7–2.

TABLE 7–2 Selected Areas under the Normal Curve

z ...	.03	.04	.05	.06
⋮	⋮	⋮	⋮	⋮
1.5	.4370	.4382	.4394	.4406
1.6	.4484	.4495	.4505	.4515
1.7	.4582	.4591	.4599	.4608
1.8	.4664	.4671	.4678	.4686

Knowing that the distance between μ and x is -1.75σ or $z = -1.75$, we can now solve for x (the minimum guaranteed mileage):

$$z = \frac{x - 67{,}900}{2{,}050}$$

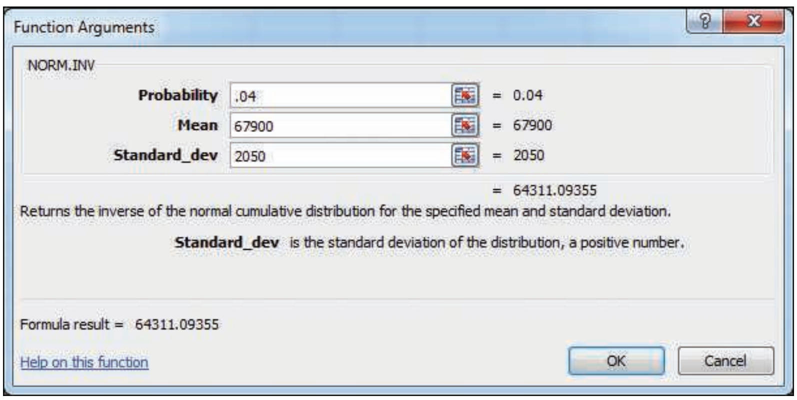
$$-1.75 = \frac{x - 67{,}900}{2{,}050}$$

$$-1.75(2{,}050) = x - 67{,}900$$

$$x = 67{,}900 - 1.75(2{,}050) = 64{,}312$$

So Layton can advertise that it will replace for free any tire that wears out before it reaches 64,312 miles, and the company will know that only 4% of the tires will be replaced under this plan.

Excel will also find the mileage value. See the following output. The necessary commands are given in the **Software Commands** in Appendix C.



SELF-REVIEW 7–6

An analysis of the final test scores for Introduction to Business reveals the scores follow the normal probability distribution. The mean of the distribution is 75 and the standard deviation is 8. The professor wants to award an A to students whose score is in the highest 10%. What is the dividing point for those students who earn an A and those earning a B?

EXERCISES

23. A normal distribution has a mean of 50 and a standard deviation of 4. Determine the value below which 95% of the observations will occur.

24. A normal distribution has a mean of 80 and a standard deviation of 14. Determine the value above which 80% of the values will occur.

25. Assume that the hourly cost to operate a commercial airplane follows the normal distribution with a mean of $2,100 per hour and a standard deviation of $250. What is the operating cost for the lowest 3% of the airplanes?

26. The SAT Reasoning Test is perhaps the most widely used standardized test for college admissions in the United States. Scores are based on a normal distribution with a mean of 1500 and a standard deviation of 300. Clinton College would like to offer an honors scholarship to students who score in the top 10% of this test. What is the minimum score that qualifies for the scholarship?

27. According to media research, the typical American listened to 195 hours of music in the last year. This is down from 290 hours 4 years earlier. Dick Trythall is a big country and western music fan. He listens to music while working around the house, reading, and riding in his truck. Assume the number of hours spent listening to music follows a normal probability distribution with a standard deviation of 8.5 hours.

a. If Dick is in the top 1% in terms of listening time, how many hours did he listen last year?

b. Assume that the distribution of times 4 years earlier also follows the normal probability distribution with a standard deviation of 8.5 hours. How many hours did the 1% who listen to the *least* music actually listen?

28. For the most recent year available, the mean annual cost to attend a private university in the United States was $42,224. Assume the distribution of annual costs follows the normal probability distribution and the standard deviation is $4,500. Ninety-five percent of all students at private universities pay less than what amount?

29. In economic theory, a "hurdle rate" is the minimum return that a person requires before he or she will make an investment. A research report says that annual returns from a specific class of common equities are distributed according to a normal distribution with a mean of 12% and a standard deviation of 18%. A stock screener would like to identify a hurdle rate such that only 1 in 20 equities is above that value. Where should the hurdle rate be set?

30. The manufacturer of a laser printer reports the mean number of pages a cartridge will print before it needs replacing is 12,200. The distribution of pages printed per cartridge closely follows the normal probability distribution and the standard deviation is 820 pages. The manufacturer wants to provide guidelines to potential customers as to how long they can expect a cartridge to last. How many pages should the manufacturer advertise for each cartridge if it wants to be correct 99% of the time?

LO7-4
Approximate the binomial probability distribution using the standard normal probability distribution to calculate probabilities.

THE NORMAL APPROXIMATION TO THE BINOMIAL

Chapter 6 describes the binomial probability distribution, which is a discrete distribution. The table of binomial probabilities in Appendix B.1 goes successively from an n of 1 to an n of 15. If a problem involved taking a sample of 60, generating a binomial distribution for that large a number would be very time-consuming. A more efficient approach is to apply the *normal approximation to the binomial*.

We can use the normal distribution (a continuous distribution) as a substitute for a binomial distribution (a discrete distribution) for large values of n because, as n increases, a binomial distribution gets closer and closer to a normal distribution. Chart 7–7 depicts the change in the shape of a binomial distribution with $\pi = .50$ from

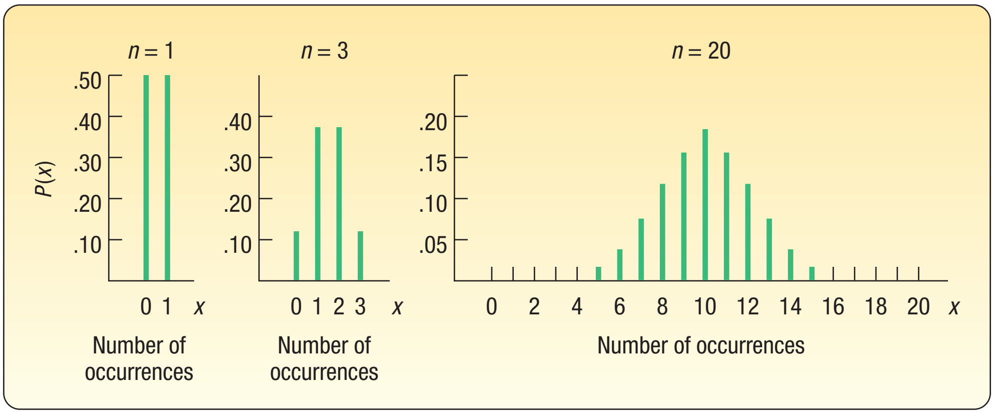


CHART 7–7 Binomial Distributions for an n of 1, 3, and 20, Where $\pi = .50$

an n of 1, to an n of 3, to an n of 20. Notice how the case where $n = 20$ approximates the shape of the normal distribution.

When can we use the normal approximation to the binomial? The normal probability distribution is a good approximation to the binomial probability distribution when $n\pi$ and $n(1 - \pi)$ are both at least 5. However, before we apply the normal approximation, we must make sure that our distribution of interest is in fact a binomial distribution. Recall from Chapter 6 that four criteria must be met:

1. There are only two mutually exclusive outcomes to an experiment: a "success" and a "failure."
2. The distribution results from counting the number of successes in a fixed number of trials.
3. The probability of a success, π, remains the same from trial to trial.
4. Each trial is independent.

Continuity Correction Factor

To show the application of the normal approximation to the binomial and the need for a correction factor, suppose the management of the Santoni Pizza Restaurant found that 70% of its new customers return for another meal. For a week in which 80 new (first-time) customers dined at Santoni's, what is the probability that 60 or more will return for another meal?

Notice the binomial conditions are met: (1) There are only two possible outcomes—a customer either returns for another meal or does not return. (2) We can count the number of successes, meaning, for example, that 57 of the 80 customers return. (3) The trials are independent, meaning that if the 34th person returns for a second meal, that does not affect whether the 58th person returns. (4) The probability of a customer returning remains at .70 for all 80 customers.

Therefore, we could use the binomial formula (6–3) described on page 185.

$$P(x) = {}_nC_x\,(\pi)^x\,(1 - \pi)^{n-x}$$

To find the probability 60 or more customers return for another pizza, we need to first find the probability exactly 60 customers return. That is:

$$P(x = 60) = {}_{80}C_{60}\,(.70)^{60}\,(1 - .70)^{20} = .063$$

Next we find the probability that exactly 61 customers return. It is:

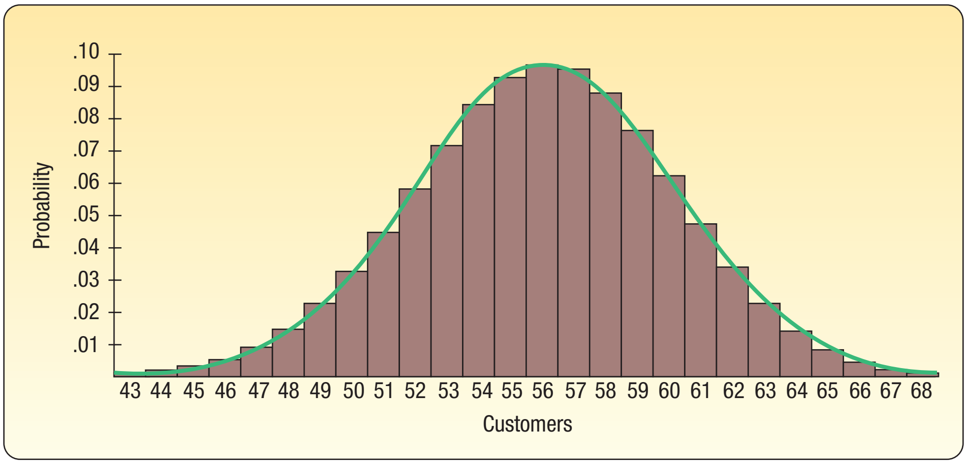
$$P(x = 61) = {}_{80}C_{61}\,(.70)^{61}\,(1 - .70)^{19} = .048$$

We continue this process until we have the probability that all 80 customers return. Finally, we add the probabilities from 60 to 80. Solving the preceding problem in this manner is tedious. We can also use statistical software packages to find the various probabilities. Listed below are the binomial probabilities for $n = 80$, $\pi = .70$, and x, the number of customers returning, ranging from 43 to 68. The probability of any number of customers less than 43 or more than 68 returning is less than .001. We can assume these probabilities are 0.000.

Number Returning	Probability	Number Returning	Probability
43	.001	56	.097
44	.002	57	.095
45	.003	58	.088
46	.006	59	.077
47	.009	60	.063
48	.015	61	.048
49	.023	62	.034
50	.033	63	.023
51	.045	64	.014
52	.059	65	.008
53	.072	66	.004
54	.084	67	.002
55	.093	68	.001

We can find the probability of 60 or more returning by summing .063 + .048 + . . . + .001, which is .197. However, a look at the plot below shows the similarity of this distribution to a normal distribution. All we need do is "smooth out" the discrete probabilities into a continuous distribution. Furthermore, working with a normal distribution will involve far fewer calculations than working with the binomial.

The trick is to let the discrete probability for 56 customers be represented by an area under the continuous curve between 55.5 and 56.5. Then let the probability for 57 customers be represented by an area between 56.5 and 57.5, and so on. This is just the opposite of rounding off the numbers to a whole number.



Because we use the normal distribution to determine the binomial probability of 60 or more successes, we must subtract, in this case, .5 from 60. The value .5 is called the **continuity correction factor.** This small adjustment is made because a continuous distribution (the normal distribution) is being used to approximate a discrete distribution (the binomial distribution).

CONTINUITY CORRECTION FACTOR The value .5 subtracted or added, depending on the question, to a selected value when a discrete probability distribution is approximated by a continuous probability distribution.

How to Apply the Correction Factor

Only four cases may arise. These cases are:

1. For the probability *at least* x occur, use the area *above* (x − .5).
2. For the probability that *more than* x occur, use the area *above* (x + .5).
3. For the probability that x *or fewer* occur, use the area *below* (x + .5).
4. For the probability that *fewer than* x occur, use the area *below* (x − .5).

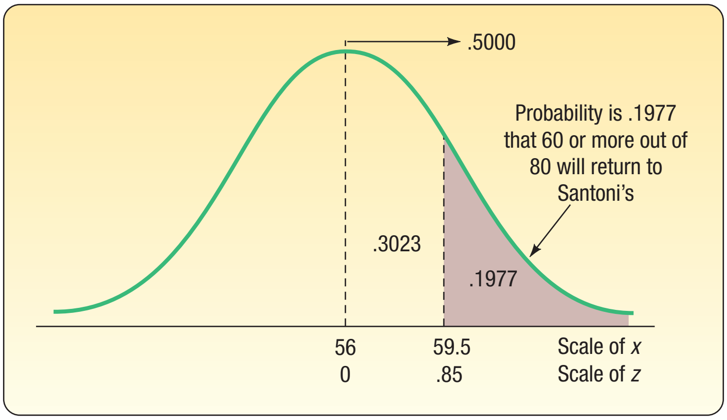
To use the normal distribution to approximate the probability that 60 or more first-time Santoni customers out of 80 will return, follow the procedure shown below.

Step 1: Find the z value corresponding to an x of 59.5 using formula (7–5), and formulas (6–4) and (6–5) for the mean and the variance of a binomial distribution:

$$\mu = n\pi = 80(.70) = 56$$
$$\sigma^2 = n\pi(1 - \pi) = 80(.70)(1 - .70) = 16.8$$
$$\sigma = \sqrt{16.8} = 4.10$$
$$z = \frac{x - \mu}{\sigma} = \frac{59.5 - 56}{4.10} = 0.85$$

Step 2: Determine the area under the normal curve between a μ of 56 and an x of 59.5. From step 1, we know that the z value corresponding to 59.5 is 0.85. So we go to Appendix B.3 and read down the left margin to 0.8, and then we go horizontally to the area under the column headed by .05. That area is .3023.

Step 3: Calculate the area beyond 59.5 by subtracting .3023 from .5000 (.5000 − .3023 = .1977). Thus, .1977 is the probability that 60 or more first-time Santoni customers out of 80 will return for another meal. In probability notation, P(customers > 59.5) = .5000 − .3023 = .1977. The facets of this problem are shown graphically:



No doubt you will agree that using the normal approximation to the binomial is a more efficient method of estimating the probability of 60 or more first-time customers returning. The result compares favorably with that computed on page 230 using the binomial distribution. The probability using the binomial distribution is .197, whereas the probability using the normal approximation is .1977.

SELF-REVIEW 7–7

A study by Great Southern Home Insurance revealed that none of the stolen goods were recovered by the homeowners in 80% of reported thefts.

(a) During a period in which 200 thefts occurred, what is the probability that no stolen goods were recovered in 170 or more of the robberies?

(b) During a period in which 200 thefts occurred, what is the probability that no stolen goods were recovered in 150 or more robberies?

EXERCISES

31. Assume a binomial probability distribution with $n = 50$ and $\pi = .25$. Compute the following:
   a. The mean and standard deviation of the random variable.
   b. The probability that x is 15 or more.
   c. The probability that x is 10 or less.

32. Assume a binomial probability distribution with $n = 40$ and $\pi = .55$. Compute the following:
   a. The mean and standard deviation of the random variable.
   b. The probability that x is 25 or greater.
   c. The probability that x is 15 or less.
   d. The probability that x is between 15 and 25, inclusive.

33. Dottie's Tax Service specializes in federal tax returns for professional clients, such as physicians, dentists, accountants, and lawyers. A recent audit by the IRS of the returns she prepared indicated that an error was made on 7% of the returns she prepared last year. Assuming this rate continues into this year and she prepares 80 returns, what is the probability that she makes errors on:
   a. More than six returns?
   b. At least six returns?
   c. Exactly six returns?

34. Shorty's Muffler advertises it can install a new muffler in 30 minutes or less. However, the work standards department at corporate headquarters recently conducted a study and found that 20% of the mufflers were not installed in 30 minutes or less. The Maumee branch installed 50 mufflers last month. If the corporate report is correct:
   a. How many of the installations at the Maumee branch would you expect to take more than 30 minutes?
   b. What is the likelihood that fewer than eight installations took more than 30 minutes?
   c. What is the likelihood that eight or fewer installations took more than 30 minutes?
   d. What is the likelihood that exactly 8 of the 50 installations took more than 30 minutes?

35. A study conducted by the nationally known Taurus Health Club revealed that 30% of its new members are 15 pounds overweight. A membership drive in a metropolitan area resulted in 500 new members.
   a. It has been suggested that the normal approximation to the binomial be used to determine the probability that 175 or more of the new members are 15 pounds overweight. Does this problem qualify as a binomial problem? Explain.
   b. What is the probability that 175 or more of the new members are 15 pounds overweight?
   c. What is the probability that 140 or more new members are 15 pounds overweight?

36. The website, **herecomestheguide.com**, suggested that couples planning their wedding should expect eighty percent of those who are sent an invitation to respond that they will attend. Rich and Stacy are planning to be married later this year. They plan to send 200 invitations.
   a. How many guests would you expect to accept the invitation?
   b. What is the standard deviation?
   c. What is the probability 150 or more will accept the invitation?
   d. What is the probability exactly 150 will accept the invitation?

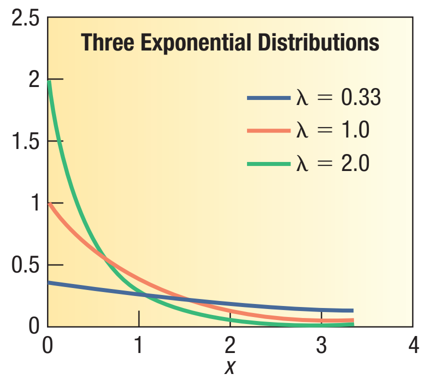
LO7-5
Describe the exponential probability distribution and use it to calculate probabilities.

THE FAMILY OF EXPONENTIAL DISTRIBUTIONS

So far in this chapter, we have considered two continuous probability distributions, the uniform and the normal. The next continuous distribution we consider is the exponential distribution. This continuous probability distribution usually describes times between events in a sequence. The actions occur independently at a constant rate per unit of time or length. Because time is never negative, an exponential random variable is always positive. The exponential distribution usually describes situations such as:

- The service time for customers at the information desk of the Dallas Public Library.
- The time between "hits" on a website.
- The lifetime of a kitchen appliance.
- The time until the next phone call arrives in a customer service center.

The exponential probability distribution is positively skewed. That differs from the uniform and normal distributions, which were both symmetric. Moreover, the distribution is described by only one parameter, which we will identify as λ (pronounced "lambda"). λ is often referred to as the "rate" parameter. The following chart shows the change in the shape of the exponential distribution as we vary the value of λ from 1/3 to 1 to 2. Observe that as we decrease λ, the shape of the distribution is "less skewed."

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Another feature of the exponential distribution is its close relationship to the Poisson distribution. The Poisson is a discrete probability distribution and also has a single parameter, μ. We described the Poisson distribution starting in Chapter 6 on page 197. It too is a positively skewed distribution. To explain the relationship between the Poisson and the exponential distributions, suppose customers arrive at a family restaurant during the dinner hour at a rate of six per hour. The Poisson distribution would have a mean of six. For a time interval of one hour, we can use the Poisson distribution to find the probability that one, or two, or ten customers arrive. But suppose instead of studying the number of customers *arriving in an hour,* we wish to study the time *between their arrivals*. The time between arrivals is a continuous distribution because time is measured as a continuous random variable. If customers arrive at a rate of six per hour, then logically the typical or mean time between arrivals is 1/6 of an hour, or 10 minutes. We need to be careful here to be consistent with our units, so let's stay with 1/6 of an hour. So in general, if we know customers arrive at a certain rate per hour, which we call μ, then we can expect the mean time between arrivals to be $1/\mu$. The rate parameter λ is equal to $1/\mu$. So in our restaurant arrival example, the mean time between customer arrivals is $\lambda = 1/6$ of an hour.

The graph of the exponential distribution starts at the value of λ when the random variable's (x) value is 0. The distribution declines steadily as we move to the right with increasing values of x. Formula (7–6) describes the exponential probability distribution with λ as rate parameter. As we described with the Poisson distribution on page 197,

e is a mathematical constant equal to 2.71828. It is the base for the natural logarithm system. It is a pleasant surprise that both the mean and the standard deviation of the exponential probability distribution are equal to $1/\lambda$.

EXPONENTIAL DISTRIBUTION $P(x) = \lambda e^{-\lambda x}$ **(7–6)**

With continuous distributions, we do not address the probability that a distinct value will occur. Instead, areas or regions below the graph of the probability distribution between two specified values give the probability the random variable is in that interval. A table, such as Appendix B.3 for the normal distribution, is not necessary for the exponential distribution. The area under the exponential density function is found by a formula and the necessary calculations can be accomplished with a handheld calculator with an e^x key. Most statistical software packages will also calculate exponential probabilities by inputting the rate parameter, λ, only. The probability of obtaining an arrival value less than a particular value of x is:

FINDING A PROBABILITY USING THE EXPONENTIAL DISTRIBUTION $P(\text{Arrival time} < x) = 1 - e^{-\lambda x}$ **(7–7)**

EXAMPLE

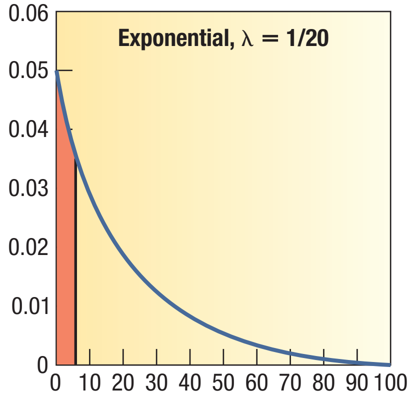
Orders for prescriptions arrive at a pharmacy website according to an exponential probability distribution at a mean of one every 20 seconds. Find the probability the next order arrives in less than 5 seconds. Find the probability the next order arrives in more than 40 seconds.

SOLUTION

To begin, we determine the rate parameter λ, which in this case is 1/20. To find the probability, we insert 1/20 for λ and 5 for x in formula (7–7).

$$P(\text{Arrival time} < 5) = 1 - e^{-\frac{1}{20}(5)} = 1 - e^{-0.25} = 1 - .7788 = .2212$$

So we conclude there is a 22% chance the next order will arrive in less than 5 seconds. The region is identified as the colored area under the curve.



The preceding computations addressed the area in the left-tail area of the exponential distribution with $\lambda = 1/20$ and the area between 0 and 5—that is, the area that is below 5 seconds. What if you are interested in the right-tail area? It is found using the complement rule. See formula (5–3) in Chapter 5. To put it another way, to find the probability the next order will arrive in more than 40 seconds, we find the probability the order arrives in less than 40 seconds and subtract the result from 1.00. We show this in two steps.

1. Find the probability an order is received *in less than* 40 seconds.

$$P(\text{Arrival} < 40) = 1 - e^{-\frac{1}{20}(40)} = 1 - .1353 = .8647$$

2. Find the probability an order is received *in more than* 40 seconds.

$$P(\text{Arrival} > 40) = 1 - P(\text{Arrival} < 40) = 1 - .8647 = .1353$$

We conclude that the likelihood that it will be 40 seconds or more before the next order is received at the pharmacy is 13.5%.

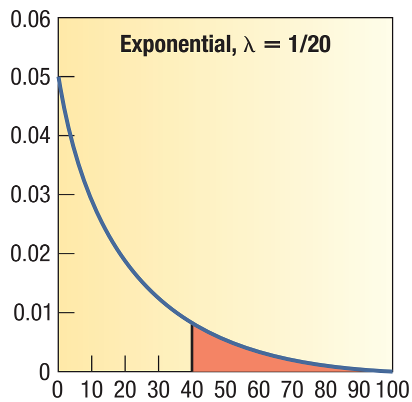
In the preceding example/solution, when we apply the exponential probability distribution to compute the probability that the arrival time is greater than 40 seconds, you probably observed that there is some redundancy. In general, if we wish to find the likelihood of a time greater than some value x, such as 40, the complement rule is applied as follows:

$$P(\text{Arrival} > x) = 1 - P(\text{Arrival} < x) = 1 - (1 - e^{-\lambda x}) = e^{-\lambda x}$$

In other words, when we subtract formula (7–7) from 1 to find the area in the right tail, the result is $e^{-\lambda x}$. Thus, the probability that more than 40 seconds go by before the next order arrives is computed without the aid of the complement rule as follows:

$$P(\text{Arrival} > 40) = e^{-\frac{1}{20}(40)} = .1353$$

The result is shown in the following graph.

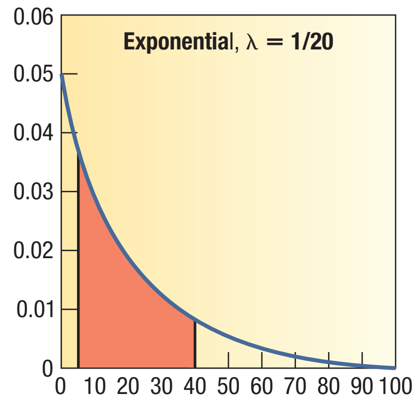


What if you wish to determine the probability that it will take more than 5 seconds but less than 40 seconds for the next order to arrive? Use formula (7–7) with an x value of 40 and then subtract the value of formula (7–7) when x is 5.

In symbols, you can write this as:

$$P(5 \le x \le 40) = P(\text{Arrival} \le 40) - P(\text{Arrival} \le 5)$$
$$= \left(1 - e^{-\frac{1}{20}(40)}\right) - \left(1 - e^{-\frac{1}{20}(5)}\right) = .8647 - .2212 = .6435$$

We conclude that about 64% of the time, the time between orders will be between 5 seconds and 40 seconds.



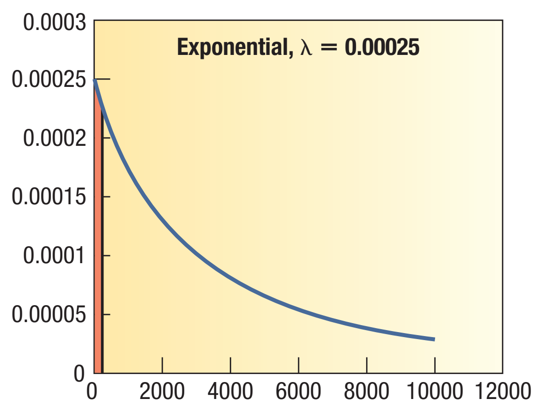
Previous examples require finding the percentage of the observations located between two values or the percentage of the observations above or below a particular value, x. We can also use formula (7–7) in "reverse" to find the value of the observation x when the percentage above or below the observation is given. The following example/solution illustrates this situation.

EXAMPLE

Compton Computers wishes to set a minimum lifetime guarantee on its new power supply unit. Quality testing shows the time to failure follows an exponential distribution with a mean of 4,000 hours. Compton wants a warranty period such that only 5% of the power supply units fail during that period. What value should they set for the warranty period?

SOLUTION

Note that 4,000 hours is a mean and not a rate. Therefore, we must compute λ as 1/4,000, or 0.00025 failure per hour. A diagram of the situation is shown below, where x represents the minimum guaranteed lifetime.



We use formula (7–7) and essentially work backward for the solution. In this case, the rate parameter is 4,000 hours and we want the area, as shown in the diagram, to be .05.

$$P(\text{Arrival time} < x) = 1 - e^{(-\lambda x)}$$

$$= 1 - e^{-\frac{1}{4{,}000}(x)} = .05$$

Next, we solve this equation for x. So, we subtract 1 from both sides of the equation and multiply by -1 to simplify the signs. The result is:

$$.95 = e^{-\frac{1}{4{,}000}(x)}$$

Next, we take the natural log of both sides and solve for x:

$$\ln(.95) = -\frac{1}{4{,}000}x$$

$$-(.051293294) = -\frac{1}{4{,}000}x$$

$$x = 205.17$$

In this case, $x = 205.17$. Hence, Compton can set the warranty period at 205 hours and expect about 5% of the power supply units to be returned.

SELF-REVIEW 7–8

The time between ambulance arrivals at the Methodist Hospital emergency room follows an exponential distribution with a mean of 10 minutes.

(a) What is the likelihood the next ambulance will arrive in 15 minutes or less?
(b) What is the likelihood the next ambulance will arrive in more than 25 minutes?
(c) What is the likelihood the next ambulance will arrive in more than 15 minutes but less than 25?
(d) Find the 80th percentile for the time between ambulance arrivals. (This means only 20% of the runs are longer than this time.)

EXERCISES

37. Waiting times to receive food after placing an order at the local Subway sandwich shop follow an exponential distribution with a mean of 60 seconds. Calculate the probability a customer waits:
   a. Less than 30 seconds.
   b. More than 120 seconds.
   c. Between 45 and 75 seconds.
   d. Fifty percent of the patrons wait less than how many seconds? What is the median?

38. The lifetime of LCD TV sets follows an exponential distribution with a mean of 100,000 hours. Compute the probability a television set:
   a. Fails in less than 10,000 hours.
   b. Lasts more than 120,000 hours.
   c. Fails between 60,000 and 100,000 hours of use.
   d. Find the 90th percentile. So 10% of the TV sets last more than what length of time?

39. The Bureau of Labor Statistics' *American Time Use Survey*, **www.bls.gov/data**, showed that the amount of time spent using a computer for leisure varied greatly by age. Individuals age 75 and over averaged 0.3 hour (18 minutes) per day using a computer for leisure. Individuals ages 15 to 19 spend 1.0 hour per day using a computer for leisure. If these times follow an exponential distribution, find the proportion of each group that spends:
   a. Less than 15 minutes per day using a computer for leisure.
   b. More than 2 hours.
   c. Between 30 minutes and 90 minutes using a computer for leisure.
   d. Find the 20th percentile. Eighty percent spend more than what amount of time?

40. The cost per item at a supermarket follows an exponential distribution. There are many inexpensive items and a few relatively expensive ones. The mean cost per item is \$3.50. What is the percentage of items that cost:
   a. Less than \$1?
   b. More than \$4?
   c. Between \$2 and \$3?
   d. Find the 40th percentile. Sixty percent of the supermarket items cost more than what amount?

CHAPTER SUMMARY

I. The uniform distribution is a continuous probability distribution with the following characteristics.
   A. It is rectangular in shape.
   B. The mean and the median are equal.
   C. It is completely described by its minimum value *a* and its maximum value *b*.
   D. It is described by the following equation for the region from *a* to *b*:

$$P(x) = \frac{1}{b - a} \tag{7–3}$$

   E. The mean and standard deviation of a uniform distribution are computed as follows:

$$\mu = \frac{(a + b)}{2} \tag{7–1}$$

$$\sigma = \sqrt{\frac{(b - a)^2}{12}} \tag{7–2}$$

II. The normal probability distribution is a continuous distribution with the following characteristics.
   A. It is bell-shaped and has a single peak at the center of the distribution.
   B. The distribution is symmetric.
   C. It is asymptotic, meaning the curve approaches but never touches the *X*-axis.
   D. It is completely described by its mean and standard deviation.
   E. There is a family of normal probability distributions.
      1. Another normal probability distribution is created when either the mean or the standard deviation changes.
      2. The normal probability distribution is described by the following formula:

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\left[\frac{(x-\mu)^2}{2\sigma^2}\right]} \tag{7–4}$$

III. The standard normal probability distribution is a particular normal distribution.
   A. It has a mean of 0 and a standard deviation of 1.
   B. Any normal probability distribution can be converted to the standard normal probability distribution by the following formula.

$$z = \frac{x - \mu}{\sigma} \tag{7–5}$$

   C. By standardizing a normal probability distribution, we can report the distance of a value from the mean in units of the standard deviation.

IV. The normal probability distribution can approximate a binomial distribution under certain conditions.
   A. $n\pi$ and $n(1 - \pi)$ must both be at least 5.
      1. n is the number of observations.
      2. π is the probability of a success.
   B. The four conditions for a binomial probability distribution are:
      1. There are only two possible outcomes.
      2. π (pi) remains the same from trial to trial.
      3. The trials are independent.
      4. The distribution results from a count of the number of successes in a fixed number of trials.
   C. The mean and variance of a binomial distribution are computed as follows:

$$\mu = n\pi$$
$$\sigma^2 = n\pi(1 - \pi)$$

   D. The continuity correction factor of .5 is used to extend the continuous value of *x* one-half unit in either direction. This correction compensates for approximating a discrete distribution by a continuous distribution.

V. The exponential probability distribution describes times between events in a sequence.
A. The actions occur independently at a constant rate per unit of time or length.
B. The probabilities are computed using the formula:

$$P(x) = \lambda e^{-\lambda x} \quad \textbf{(7–6)}$$

C. It is nonnegative, is positively skewed, declines steadily to the right, and is asymptotic.
D. The area under the curve is given by the formula

$$P(\text{Arrival time} < x) = 1 - e^{-\lambda x} \quad \textbf{(7–7)}$$

E. Both the mean and standard deviation are:

$$\mu = 1/\lambda$$
$$\sigma^2 = 1/\lambda$$

CHAPTER EXERCISES

41. The amount of cola in a 12-ounce can is uniformly distributed between 11.96 ounces and 12.05 ounces.
- **a.** What is the mean amount per can?
- **b.** What is the standard deviation amount per can?
- **c.** What is the probability of selecting a can of cola and finding it has less than 12 ounces?
- **d.** What is the probability of selecting a can of cola and finding it has more than 11.98 ounces?
- **e.** What is the probability of selecting a can of cola and finding it has more than 11.00 ounces?

42. A tube of Listerine Tartar Control toothpaste contains 4.2 ounces. As people use the toothpaste, the amount remaining in any tube is random. Assume the amount of toothpaste remaining in the tube follows a uniform distribution. From this information, we can determine the following information about the amount remaining in a toothpaste tube without invading anyone's privacy.
- **a.** How much toothpaste would you expect to be remaining in the tube?
- **b.** What is the standard deviation of the amount remaining in the tube?
- **c.** What is the likelihood there is less than 3.0 ounces remaining in the tube?
- **d.** What is the probability there is more than 1.5 ounces remaining in the tube?

43. Many retail stores offer their own credit cards. At the time of the credit application, the customer is given a 10% discount on the purchase. The time required for the credit application process follows a uniform distribution with the times ranging from 4 minutes to 10 minutes.
- **a.** What is the mean time for the application process?
- **b.** What is the standard deviation of the process time?
- **c.** What is the likelihood a particular application will take less than 6 minutes?
- **d.** What is the likelihood an application will take more than 5 minutes?

44. The time patrons at the Grande Dunes Hotel in the Bahamas spend waiting for an elevator follows a uniform distribution between 0 and 3.5 minutes.
- **a.** Show that the area under the curve is 1.00.
- **b.** How long does the typical patron wait for elevator service?
- **c.** What is the standard deviation of the waiting time?
- **d.** What percent of the patrons wait for less than a minute?
- **e.** What percent of the patrons wait more than 2 minutes?

45. The net sales and the number of employees for aluminum fabricators with similar characteristics are organized into frequency distributions. Both are normally distributed. For the net sales, the mean is $180 million and the standard deviation is $25 million. For the number of employees, the mean is 1,500 and the standard deviation is 120. Clarion Fabricators had sales of $170 million and 1,850 employees.
- **a.** Convert Clarion's sales and number of employees to *z* values.
- **b.** Locate the two *z* values.
- **c.** Compare Clarion's sales and number of employees with those of the other fabricators.

46. The accounting department at Weston Materials Inc., a national manufacturer of unattached garages, reports that it takes two construction workers a mean of 32 hours and a standard deviation of 2 hours to erect the Red Barn model. Assume the assembly times follow the normal distribution.
  a. Determine the *z* values for 29 and 34 hours. What percent of the garages take between 32 hours and 34 hours to erect?
  b. What percent of the garages take between 29 hours and 34 hours to erect?
  c. What percent of the garages take 28.7 hours or less to erect?
  d. Of the garages, 5% take how many hours or more to erect?

47. In 2015 The United States Department of Agriculture issued a report (**http://www.cnpp.usda.gov/sites/default/files/CostofFoodMar2015.pdf**) indicating a family of four spent an average of about $890 per month on food. Assume the distribution of food expenditures for a family of four follows the normal distribution, with a standard deviation of $90 per month.
  a. What percent of the families spend more than $430 but less than $890 per month on food?
  b. What percent of the families spend less than $830 per month on food?
  c. What percent spend between $830 and $1,000 per month on food?
  d. What percent spend between $900 and $1,000 per month on food?

48. A study of long-distance phone calls made from General Electric Corporate Headquarters in Fairfield, Connecticut, revealed the length of the calls, in minutes, follows the normal probability distribution. The mean length of time per call was 4.2 minutes and the standard deviation was 0.60 minute.
  a. What is the probability that calls last between 4.2 and 5 minutes?
  b. What is the probability that calls last more than 5 minutes?
  c. What is the probability that calls last between 5 and 6 minutes?
  d. What is the probability that calls last between 4 and 6 minutes?
  e. As part of her report to the president, the director of communications would like to report the length of the longest (in duration) 4% of the calls. What is this time?

49. Shaver Manufacturing Inc. offers dental insurance to its employees. A recent study by the human resource director shows the annual cost per employee per year followed the normal probability distribution, with a mean of $1,280 and a standard deviation of $420 per year.
  a. What is the probability that annual dental expenses are more than $1,500?
  b. What is the probability that annual dental expenses are between $1,500 and $2,000?
  c. Estimate the probability that an employee had no annual dental expenses.
  d. What was the cost for the 10% of employees who incurred the highest dental expense?

50. The annual commissions earned by sales representatives of Machine Products Inc., a manufacturer of light machinery, follow the normal probability distribution. The mean yearly amount earned is $40,000 and the standard deviation is $5,000.
  a. What percent of the sales representatives earn more than $42,000 per year?
  b. What percent of the sales representatives earn between $32,000 and $42,000?
  c. What percent of the sales representatives earn between $32,000 and $35,000?
  d. The sales manager wants to award the sales representatives who earn the largest commissions a bonus of $1,000. He can award a bonus to 20% of the representatives. What is the cutoff point between those who earn a bonus and those who do not?

51. According to the South Dakota Department of Health, the number of hours of TV viewing per week is higher among adult women than adult men. A recent study showed women spent an average of 34 hours per week watching TV, and men, 29 hours per week. Assume that the distribution of hours watched follows the normal distribution for both groups, and that the standard deviation among the women is 4.5 hours and is 5.1 hours for the men.
  a. What percent of the women watch TV less than 40 hours per week?
  b. What percent of the men watch TV more than 25 hours per week?
  c. How many hours of TV do the 1% of women who watch the most TV per week watch? Find the comparable value for the men.

52. According to a government study among adults in the 25- to 34-year age group, the mean amount spent per year on reading and entertainment is \$1,994. Assume that the distribution of the amounts spent follows the normal distribution with a standard deviation of \$450.

a. What percent of the adults spend more than \$2,500 per year on reading and entertainment?

b. What percent spend between \$2,500 and \$3,000 per year on reading and entertainment?

c. What percent spend less than \$1,000 per year on reading and entertainment?

53. Management at Gordon Electronics is considering adopting a bonus system to increase production. One suggestion is to pay a bonus on the highest 5% of production based on past experience. Past records indicate weekly production follows the normal distribution. The mean of this distribution is 4,000 units per week and the standard deviation is 60 units per week. If the bonus is paid on the upper 5% of production, the bonus will be paid on how many units or more?

54. Fast Service Truck Lines uses the Ford Super Duty F-750 exclusively. Management made a study of the maintenance costs and determined the number of miles traveled during the year followed the normal distribution. The mean of the distribution was 60,000 miles and the standard deviation 2,000 miles.

a. What percent of the Ford Super Duty F-750s logged 65,200 miles or more?

b. What percent of the trucks logged more than 57,060 but less than 58,280 miles?

c. What percent of the Fords traveled 62,000 miles or less during the year?

d. Is it reasonable to conclude that any of the trucks were driven more than 70,000 miles? Explain.

55. Best Electronics Inc. offers a “no hassle” returns policy. The daily number of customers returning items follows the normal distribution. The mean number of customers returning items is 10.3 per day and the standard deviation is 2.25 per day.

a. For any day, what is the probability that eight or fewer customers returned items?

b. For any day, what is the probability that the number of customers returning items is between 12 and 14?

c. Is there any chance of a day with no customer returns?

56. A recent news report indicated that 20% of all employees steal from their company each year. If a company employs 50 people, what is the probability that:

a. Fewer than five employees steal?

b. More than five employees steal?

c. Exactly five employees steal?

d. More than 5 but fewer than 15 employees steal?

57. The *Orange County Register,* as part of its Sunday health supplement, reported that 64% of American men over the age of 18 consider nutrition a top priority in their lives. Suppose we select a sample of 60 men. What is the likelihood that:

a. 32 or more consider nutrition important?

b. 44 or more consider nutrition important?

c. More than 32 but fewer than 43 consider nutrition important?

d. Exactly 44 consider diet important?

58. It is estimated that 10% of those taking the quantitative methods portion of the CPA examination fail that section. Sixty students are taking the exam this Saturday.

a. How many would you expect to fail? What is the standard deviation?

b. What is the probability that exactly two students will fail?

c. What is the probability at least two students will fail?

59. The Georgetown, South Carolina, Traffic Division reported 40% of high-speed chases involving automobiles result in a minor or major accident. If 50 high-speed chases occur in a year, what is the probability that 25 or more will result in a minor or major accident?

60. Cruise ships of the Royal Viking line report that 80% of their rooms are occupied during September. For a cruise ship having 800 rooms, what is the probability that 665 or more are occupied in September?

61. The goal at U.S. airports handling international flights is to clear these flights within 45 minutes. Let’s interpret this to mean that 95% of the flights are cleared in 45 minutes, so 5% of the flights take longer to clear. Let’s also assume that the distribution is approximately normal.

a. If the standard deviation of the time to clear an international flight is 5 minutes, what is the mean time to clear a flight?

b. Suppose the standard deviation is 10 minutes, not the 5 minutes suggested in part (a). What is the new mean?

c. A customer has 30 minutes from the time her flight lands to catch her limousine. Assuming a standard deviation of 10 minutes, what is the likelihood that she will be cleared in time?

62. The funds dispensed at the ATM machine located near the checkout line at the Kroger's in Union, Kentucky, follows a normal probability distribution with a mean of $4,200 per day and a standard deviation of $720 per day. The machine is programmed to notify the nearby bank if the amount dispensed is very low (less than $2,500) or very high (more than $6,000).

a. What percent of the days will the bank be notified because the amount dispensed is very low?

b. What percent of the time will the bank be notified because the amount dispensed is high?

c. What percent of the time will the bank not be notified regarding the amount of funds dispersed?

63. The weights of canned hams processed at Henline Ham Company follow the normal distribution, with a mean of 9.20 pounds and a standard deviation of 0.25 pound. The label weight is given as 9.00 pounds.

a. What proportion of the hams actually weigh less than the amount claimed on the label?

b. The owner, Glen Henline, is considering two proposals to reduce the proportion of hams below label weight. He can increase the mean weight to 9.25 and leave the standard deviation the same, or he can leave the mean weight at 9.20 and reduce the standard deviation from 0.25 pound to 0.15. Which change would you recommend?

64. A recent Gallup study (http://www.gallup.com/poll/175286/hour-workweek-actually-longer-seven-hours.aspx) found the typical American works an average of 46.7 hour per week. The study did not report the shape of the distribution of hours worked or the standard deviation. It did however indicate that 40% of the workers worked less than 40 hours a week and that 18 percent worked more than 60 hours.

a. If we assume that the distribution of hours worked is normally distributed, and knowing 40% of the workers worked less than 40 hours, find the standard deviation of the distribution.

b. If we assume that the distribution of hours worked is normally distributed and 18% of the workers worked more than 60 hours, find the standard deviation of the distribution.

c. Compare the standard deviations computed in parts *a* and *b*. Is the assumption that the distribution of hours worked is approximately normal reasonable? Why?

65. Most four-year automobile leases allow up to 60,000 miles. If the lessee goes beyond this amount, a penalty of 20 cents per mile is added to the lease cost. Suppose the distribution of miles driven on four-year leases follows the normal distribution. The mean is 52,000 miles and the standard deviation is 5,000 miles.

a. What percent of the leases will yield a penalty because of excess mileage?

b. If the automobile company wanted to change the terms of the lease so that 25% of the leases went over the limit, where should the new upper limit be set?

c. One definition of a low-mileage car is one that is 4 years old and has been driven less than 45,000 miles. What percent of the cars returned are considered low-mileage?

66. The price of shares of Bank of Florida at the end of trading each day for the last year followed the normal distribution. Assume there were 240 trading days in the year. The mean price was $42.00 per share and the standard deviation was $2.25 per share.

a. What is the probability that the end-of-day trading price is over $45.00? Estimate the number of days in a year when the trading price finished above $45.00.

b. What percent of the days was the price between $38.00 and $40.00?

c. What is the minimum share price for the top 15% of end-of-day trading prices?

67. The annual sales of romance novels follow the normal distribution. However, the mean and the standard deviation are unknown. Forty percent of the time sales are more than 470,000, and 10% of the time sales are more than 500,000. What are the mean and the standard deviation?

68. In establishing warranties on HDTVs, the manufacturer wants to set the limits so that few will need repair at the manufacturer's expense. On the other hand, the warranty period must be long enough to make the purchase attractive to the buyer. For a new HDTV, the mean number of months until repairs are needed is 36.84 with a standard deviation of 3.34 months. Where should the warranty limits be set so that only 10% of the HDTVs need repairs at the manufacturer's expense?

69. DeKorte Tele-Marketing Inc. is considering purchasing a machine that randomly selects and automatically dials telephone numbers. DeKorte Tele-Marketing makes most of its calls during the evening, so calls to business phones are wasted. The manufacturer of the machine claims that its programming reduces the calling to business phones to 15% of all calls. To test this claim, the director of purchasing at DeKorte programmed the machine to select a sample of 150 phone numbers. What is the likelihood that more than 30 of the phone numbers selected are those of businesses, assuming the manufacturer's claim is correct?

70. A carbon monoxide detector in the Wheelock household activates once every 200 days on average. Assume this activation follows the exponential distribution. What is the probability that:

- **a.** There will be an alarm within the next 60 days?
- **b.** At least 400 days will pass before the next alarm?
- **c.** It will be between 150 and 250 days until the next warning?
- **d.** Find the median time until the next activation.

71. "Boot time" (the time between the appearance of the Bios screen to the first file that is loaded in Windows) on Eric Mouser's personal computer follows an exponential distribution with a mean of 27 seconds. What is the probability his "boot" will require:

- **a.** Less than 15 seconds?
- **b.** More than 60 seconds?
- **c.** Between 30 and 45 seconds?
- **d.** What is the point below which only 10% of the boots occur?

72. The time between visits to a U.S. emergency room for a member of the general population follows an exponential distribution with a mean of 2.5 years. What proportion of the population:

- **a.** Will visit an emergency room within the next 6 months?
- **b.** Will not visit the ER over the next 6 years?
- **c.** Will visit an ER next year, but not this year?
- **d.** Find the first and third quartiles of this distribution.

73. The times between failures on a personal computer follow an exponential distribution with a mean of 300,000 hours. What is the probability of:

- **a.** A failure in less than 100,000 hours?
- **b.** No failure in the next 500,000 hours?
- **c.** The next failure occurring between 200,000 and 350,000 hours?
- **d.** What are the mean and standard deviation of the time between failures?

DATA ANALYTICS

(The data for these exercises are available at the text website: www.mhhe.com/lind17e.)

74. Refer to the North Valley Real Estate data, which report information on homes sold during the last year.

- **a.** The mean selling price (in $ thousands) of the homes was computed earlier to be $357.0, with a standard deviation of $160.7. Use the normal distribution to estimate the percentage of homes selling for more than $500.000. Compare this to the actual results. Is price normally distributed? Try another test. If price is normally distributed, how many homes should have a price greater than the mean? Compare this to the actual number of homes. Construct a frequency distribution of price. What do you observe?
- **b.** The mean days on the market is 30 with a standard deviation of 10 days. Use the normal distribution to estimate the number of homes on the market more than 24 days. Compare this to the actual results. Try another test. If days on the market is normally distributed, how many homes should be on the market more than the mean number of days? Compare this to the actual number of homes. Does the normal

distribution yield a good approximation of the actual results? Create a frequency distribution of days on the market. What do you observe?

75. Refer to the Baseball 2016 data, which report information on the 30 Major League Baseball teams for the 2016 season.

a. The mean attendance per team for the season was 2.439 million, with a standard deviation of 0.618 million. Use the normal distribution to estimate the number of teams with attendance of more than 3.5 million. Compare that estimate with the actual number. Comment on the accuracy of your estimate.

b. The mean team salary was $121 million, with a standard deviation of $40.0 million. Use the normal distribution to estimate the number of teams with a team salary of more than $100 million. Compare that estimate with the actual number. Comment on the accuracy of the estimate.

76. Refer to the Lincolnville School District bus data.

a. Refer to the maintenance cost variable. The mean maintenance cost for last year is $4,552 with a standard deviation of $2332. Estimate the number of buses with a maintenance cost of more than $6,000. Compare that with the actual number. Create a frequency distribution of maintenance cost. Is the distribution normally distributed?

b. Refer to the variable on the number of miles driven since the last maintenance. The mean is 11,121 and the standard deviation is 617 miles. Estimate the number of buses traveling more than 11,500 miles since the last maintenance. Compare that number with the actual value. Create a frequency distribution of miles since maintenance cost. Is the distribution normally distributed?

A REVIEW OF CHAPTERS 5–7

The chapters in this section consider methods of dealing with uncertainty. In Chapter 5, we describe the concept of probability. A *probability* is a value between 0 and 1 that expresses the likelihood a particular event will occur. We also looked at methods to calculate probabilities using rules of addition and multiplication; presented principles of counting, including permutations and combinations; and described situations for using Bayes' theorem.

Chapter 6 describes *discrete* probability distributions. Discrete probability distributions list all possible outcomes of an experiment and the probability associated with each outcome. We describe three discrete probability distributions: the *binomial distribution,* the *hypergeometric distribution,* and the *Poisson distribution.* The requirements for the binomial distribution are there are only two possible outcomes for each trial, there is a constant probability of success, there are a fixed number of trials, and the trials are independent. The binomial distribution lists the probabilities for the number of successes in a fixed number of trials. The hypergeometric distribution is similar to the binomial, but the probability of success is not constant, so the trials are not independent. The Poisson distribution is characterized by a small probability of success in a large number of trials. It has the following characteristics: the random variable is the number of times some event occurs in a fixed interval, the probability of a success is proportional to the size of the interval, and the intervals are independent and do not overlap.

Chapter 7 describes three continuous probability distributions: the *uniform distribution,* the *normal distribution,* and the *exponential distribution.* The uniform probability distribution is rectangular in shape and is defined by minimum and maximum values. The mean and the median of a uniform probability distribution are equal, and it does not have a mode.

A normal probability distribution is the most widely used and widely reported distribution. Its major characteristics are that it is bell-shaped and symmetrical, completely described by its mean and standard deviation, and asymptotic, that is, it falls smoothly in each direction from its peak but never touches the horizontal axis. There is a family of normal probability distributions—each with its own mean and standard deviation. There are an unlimited number of normal probability distributions.

To find the probabilities for any normal probability distribution, we convert a normal distribution to a *standard normal probability distribution* by computing *z* values. A *z* value is the distance between *x* and the mean in units of the standard deviation. The standard normal probability distribution has a mean of 0 and a standard deviation of 1. It is useful because the probability for any event from a normal probability distribution can be computed using standard normal probability tables (see Appendix B.3).

The exponential probability distribution describes the time between events in a sequence. These events occur independently at a constant rate per unit of time or length. The exponential probability distribution is positively skewed, with λ as the "rate" parameter. The mean and standard deviation are equal and are the reciprocal of λ.

PROBLEMS

1. Proactine, a new medicine for acne, is claimed by the manufacturer to be 80% effective. It is applied to the affected area of a sample of 15 people. What is the probability that:
   a. All 15 will show significant improvement?
   b. Fewer than 9 of 15 will show significant improvement?
   c. 12 or more people will show significant improvement?
2. Customers at the Bank of Commerce of Idaho Falls, Idaho, default at a rate of .005 on small home-improvement loans. The bank has approved 400 small home-improvement loans. Assuming the Poisson probability distribution applies to this problem:
   a. What is the probability that no homeowners out of the 400 will default?
   b. How many of the 400 are expected not to default?
   c. What is the probability that three or more homeowners will default on their small home-improvement loans?
3. A study of the attendance at the University of Alabama's basketball games revealed that the distribution of attendance is normally distributed with a mean of 10,000 and a standard deviation of 2,000.
   a. What is the probability a particular game has an attendance of 13,500 or more?
   b. What percent of the games have an attendance between 8,000 and 11,500?
   c. Ten percent of the games have an attendance of how many or less?
4. **FILE** Daniel-James Insurance Company will insure an offshore ExxonMobil oil production platform against weather losses for one year. The president of Daniel-James estimates the following losses for that platform (in millions of dollars) with the accompanying probabilities:

Amount of Loss ($ millions)	Probability of Loss
0	.98
40	.016
300	.004

   a. What is the expected amount Daniel-James will have to pay to ExxonMobil in claims?
   b. What is the likelihood that Daniel-James will actually lose less than the expected amount?
   c. Given that Daniel-James suffers a loss, what is the likelihood that it is for $300 million?
   d. Daniel-James has set the annual premium at $2.0 million. Does that seem like a fair premium? Will it cover its risk?
5. **FILE** The distribution of the number of school-age children per family in the Whitehall Estates area of Grand Junction, Colorado, is:

Number of children	0	1	2	3	4
Percent of families	40	30	15	10	5

   a. Determine the mean and standard deviation of the number of school-age children per family in Whitehall Estates.
   b. A new school is planned in Whitehall Estates. An estimate of the number of school-age children is needed. There are 500 family units. How many children would you estimate?
   c. Some additional information is needed about only the families having children. Convert the preceding distribution to one for families with children. What is the mean number of children among families that have children?
6. The following table shows a breakdown of the 114th U.S. Congress by party affiliation. (There are two independent senators included in the count of Democratic senators. There is one vacant House seat.)

	Party		
	Democrats	Republicans	Total
House	188	246	434
Senate	46	54	100
Total	234	300	534

a. A member of Congress is selected at random. What is the probability of selecting a Republican?
b. Given that the person selected is a member of the House of Representatives, what is the probability he or she is a Republican?
c. What is the probability of selecting a member of the House of Representatives or a Democrat?

CASES

A. Century National Bank

Refer to the Century National Bank data. Is it reasonable that the distribution of checking account balances approximates a normal probability distribution? Determine the mean and the standard deviation for the sample of 60 customers. Compare the actual distribution with the theoretical distribution. Cite some specific examples and comment on your findings.

Divide the account balances into three groups, of about 20 each, with the smallest third of the balances in the first group, the middle third in the second group, and those with the largest balances in the third group. Next, develop a table that shows the number in each of the categories of the account balances by branch. Does it appear that account balances are related to the branch? Cite some examples and comment on your findings.

B. Elections Auditor

An item such as an increase in taxes, recall of elected officials, or an expansion of public services can be placed on the ballot if a required number of valid signatures are collected on the petition. Unfortunately, many people will sign the petition even though they are not registered to vote in that particular district, or they will sign the petition more than once.

Sara Ferguson, the elections auditor in Venango County, must certify the validity of these signatures after the petition is officially presented. Not surprisingly, her staff is overloaded, so she is considering using statistical methods to validate the pages of 200 signatures, instead of validating each individual signature. At a recent professional meeting, she found that, in some communities in the state, election officials were checking only five signatures on each page and rejecting the entire page if two or more signatures were invalid. Some people are concerned that five may not be enough to make a good decision. They suggest that you should check 10 signatures and reject the page if 3 or more are invalid.

In order to investigate these methods, Sara asks her staff to pull the results from the last election and sample 30 pages. It happens that the staff selected 14 pages from the Avondale district, 9 pages from the Midway district, and 7 pages from the Kingston district. Each page had 200 signatures, and the data below show the number of invalid signatures on each.

Use the data to evaluate Sara's two proposals. Calculate the probability of rejecting a page under each of the approaches. Would you get about the same results by examining every single signature? Offer a plan of your own, and discuss how it might be better or worse than the two plans proposed by Sara.

Avondale	Midway	Kingston
9	19	38
14	22	39
11	23	41
8	14	39
14	22	41
6	17	39
10	15	39
13	20	
8	18	
8		
9		
12		
7		
13		

C. Geoff Applies Data Analytics

Geoff Brown is the manager for a small telemarketing firm and is evaluating the sales rate of experienced workers in order to set minimum standards for new hires. During the past few weeks, he has recorded the number of successful calls per hour for the staff. These data appear next along with some summary statistics he worked out with a statistical software package. Geoff has been a student at the local community college and has heard of many different kinds of probability distributions (binomial, normal, hypergeometric, Poisson, etc.). Could you give Geoff some advice on which distribution to use to fit these data as well as possible and how to decide when a probationary employee should be accepted as having reached full production status? This is important because it means a pay raise for the employee, and there have been some probationary employees in the past who have quit because of discouragement that they would never meet the standard.

Successful sales calls per hour during the week of August 14:

4	2	3	1	4	5	5	2	3	2	2	4	5	2	5	3	3	0
1	3	2	8	4	5	2	2	4	1	5	5	4	5	1	2	4	

Descriptive statistics:

N	MEAN	MEDIAN	STANDARD DEVIATION
35	3.229	3.000	1.682
MIN	MAX	1ST QUARTILE	3RD QUARTILE
0.0	8.0	2.0	5.0

Analyze the distribution of sales calls. Which distribution do you think Geoff should use for his analysis? Support your recommendation with your analysis. What standard should be used to determine if an employee has reached "full production" status? Explain your recommendation.

D. CNP Bank Card

Before banks issue a credit card, they usually rate or score the customer in terms of his or her projected probability of being a profitable customer. A typical scoring table appears below.

Age	**Under 25** (12 pts.)	**25–29** (5 pts.)	**30–34** (0 pts.)	**35+** (18 pts.)
Time at same address	**<1 yr.** (9 pts.)	**1–2 yrs.** (0 pts.)	**3–4 yrs.** (13 pts.)	**5+ yrs.** (20 pts.)
Auto age	**None** (18 pts.)	**0–1yr.** (12 pts.)	**2–4 yrs.** (13 pts.)	**5+ yrs.** (3 pts.)
Monthly car payment	**None** (15 pts.)	**$1–$99** (6 pts.)	**$100–$299** (4 pts.)	**$300+** (0 pts.)
Housing cost	**$1–$199** (0 pts.)	**$200–$399** (10 pts.)	**Owns** (12 pts.)	**Lives with relatives** (24 pts.)
Checking/ savings accounts	**Both** (15 pts.)	**Checking only** (3 pts.)	**Savings only** (2 pts.)	**Neither** (0 pts.)

The score is the sum of the points on the six items. For example, Tracy Brown is under 25 years old (12 pts.), has lived at the same address for 2 years (0 pts.), owns a 4-year-old car (13 pts.), with car payments of $75 (6 pts.), housing cost of $450 (10 pts.), and a checking account (3 pts.). She would score 44.

A second chart is then used to convert scores into the probability of being a profitable customer. A sample chart of this type appears below.

Score	30	40	50	60	70	80	90
Probability	.70	.78	.85	.90	.94	.95	.96

Tracy's score of 44 would translate into a probability of being profitable of approximately .81. In other words, 81% of customers like Tracy will make money for the bank card operations.

Here are the interview results for three potential customers.

Name	David Born	Edward Brendan	Ann McLaughlin
Age	42	23	33
Time at same address	9	2	5
Auto age	2	3	7
Monthly car payment	$140	$99	$175
Housing cost	$450	$650	Owns clear
Checking/savings accounts	Both	Checking only	Neither

1. Score each of these customers and estimate their probability of being profitable.
2. What is the probability that all three are profitable?
3. What is the probability that none of them are profitable?
4. Find the entire probability distribution for the number of profitable customers among this group of three.
5. Write a brief summary of your findings.

PRACTICE TEST

Part 1—Objective

1. Under what conditions will a probability be greater than 1 or 100%? **1.** ________
2. An ________ is the observation of some activity or the act of taking some type of measurement. **2.** ________
3. An ________ is the collection of one or more outcomes to an experiment. **3.** ________
4. A ________ probability is the likelihood that two or more events will happen at the same time. **4.** ________
5. In a (5a) ________, the order in which the events are counted is important, but in a (5b) ________, it is not important. **5. a.** ________ **5. b.** ________
6. In a discrete probability distribution, the sum of the possible outcomes is equal to ________. **6.** ________
7. Which of the following is *NOT* a requirement of the binomial distribution? (constant probability of success, three or more outcomes, the result of counts) **7.** ________
8. How many normal distributions are there? (1, 10, 30, 1,000, or infinite—pick one) **8.** ________
9. How many standard normal distributions are there? (1, 10, 30, 1,000, or infinite—pick one) **9.** ________
10. What is the probability of finding a z value between 0 and −0.76? **10.** ________
11. What is the probability of finding a z value greater than 1.67? **11.** ________
12. Two events are ____________ if the occurrence of one event does not affect the occurrence of another event. **12.** ________

13. Two events are ____________ if by virtue of one event happening the other cannot happen. **13.** ____________

14. Which of the following is not true regarding the normal probability distribution? (asymptotic, family of distributions, only two outcomes, 50% of the observations greater than the mean) **14.** ____________

15. Which of the following statements best describes the shape of a normal probability distribution? (bell-shaped, uniform, V-shaped, no constant shape) **15.** ____________

Part 2—Problems

1. Fred Friendly, CPA, has 20 tax returns to prepare before the April 15th deadline. It is late at night so he decides to do two more before going home. In his stack of accounts, 12 are personal, 5 are businesses, and 3 are for charitable organizations. If he selects the two returns at random, what is the probability:

a. Both are businesses?
b. At least one is a business?

2. The IRS reports that 15% of returns where the adjusted gross income is more than $1,000,000 will be subject to a computer audit. For the year 2017, Fred Friendly, CPA, completed 16 returns where the adjusted gross income was more than $1,000,000.

a. What is the probability exactly one of these returns will be audited?
b. What is the probability at least one will be audited?

3. Fred works in a tax office with five other CPAs. There are five parking spots beside the office. In how many different ways can the cars belonging to the CPAs be arranged in the five spots? Assume they all drive to work.

4. Fred decided to study the number of exemptions claimed on personal tax returns he prepared in 2017. The data are summarized in the following table.

Exemptions	Percent
1	20
2	50
3	20
4	10

a. What is the mean number of exemptions per return?
b. What is the variance of the number of exemptions per return?

5. In a memo to all those involved in tax preparation, the IRS indicated that the mean amount of refund was $1,600 with a standard deviation of $850. Assume the distribution of the amounts returned follows the normal distribution.

a. What percent of the refunds were between $1,600 and $2,000?
b. What percent of the refunds were between $900 and $2,000?
c. According to the above information, what percent of the refunds were less than $0; that is, the taxpayer owed the IRS.

6. For the year 2017, Fred Friendly completed a total of 80 returns. He developed the following table summarizing the relationship between number of dependents and whether or not the client received a refund.

	Dependents			
Refund	**1**	**2**	**3 or more**	**Total**
Yes	20	20	10	50
No	10	20	0	30
Total	30	40	10	80

a. What is the name given to this table?
b. What is the probability of selecting a client who received a refund?
c. What is the probability of selecting a client who received a refund or had one dependent?
d. Given that the client received a refund, what is the probability he or she had one dependent?
e. What is the probability of selecting a client who did *not* receive a refund and had one dependent?

7. The IRS offers taxpayers the choice of allowing the IRS to compute the amount of their tax refund. During the busy filing season, the number of returns received at the Springfield Service Center that request this service follows a Poisson distribution with a mean of three per day. What is the probability that on a particular day:

a. There are no requests?
b. Exactly three requests appear?
c. Five or more requests take place?
d. There are no requests on two consecutive days?